Diagnostic Proposal Document for CHANA CD101

长安汽车

Version 2.7

2012-03-07

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Change History

Rev	Date	Section	Change description	Author
1.0	28.03.2010	All	Initial for proposal to CHANA C201 (Draft)	Liang Xie
1.1	30.03.2010	2	Add the supported Routines control	Liang Xie
1.2	12.05.2010	All	Diagnostic Annex file for CHANA C201 proposal	Liang Xie
1.3	28.05.2010	3.3, 4.1	Changed some DID and DTC as per CHANA request.	Liang Xie
1.4	30.05.2010	5	Add the detailed diagnostic services data definition from CHANA	Liang Xie
1,5	20.06.2010	2	Change some sub-functions as discussed with CHANA.	Liang Xie
1.6	24.06.2010	5	Add all the content of CHANA UDS diagnostic requirement specification	Liang Xie
1.8	20.04.2011	All	Update the DTC list, Car configuration, EOL diagnostic session, Diagnostic service conditions and so on for CHANA CD101 ABS/VSC.	Liang Xie
2.0	14.06.2011	3.4	Delete standard diagnostic session for car configuration; Change the car configuration logic values to match CAN message map definition;	Liang XIe
2.0	14.06.2011	3.5.3	Add EOL test by routine control	Liang Xie
2.0	14.06.2011	6.1.11	Change the I/O control vehicle speed limitation value from 10 KPH to 3 KPH..	Liang Xie
2.0	14.06.2011	4.1	Update the DTC list by adding the DTCs related to CD101 EPB function.	Liang Xie
2.2	30.10.2011	All	Change some items with comments.	Liang Xie
2.2	30.10.2011	3.6.1	Add the routine control results response for CHANA projects.	Liang Xie
2.3	2011-11-17	All	Rearrange all the sessions	Jie Gong
2.4	2012-1-18	3.3.10	Add the note for current implementation of the snap shot data record number	Liu Feng
		4.1	DID 3001 – FS relay bit is marked as short term adjustment is not allowed	
		4.4.	Update the configuration data and variant coding description	

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		4.6.1	1,Correct the request of routine control request to $31 $03 $XX $YY instead of $31 $XX $YY $03 2,Update the description of sensor calibration fails 3, Add 'X' mark in the StopRoutine (02) column for the IDs $3020, $3021, $3027 indicating that stop is not supported for this routines in Table 3.7 Supported Routines in application SW.	
		5.1	Update the DTC list per CD101 implementation	
		5.2.4	Updated the information in table 5-4	
2.5	2012-1-21	3.2	1, Freeze current state sub function is removed from IO control service. 2, "Only in Programming session" comment has to be given against Transfer Data (36) and Request TransferExit (37) services	Liu Feng
		5.1	1,Updated the description of DTC 223 & 224 2,Remove fault of ‘ EPB Clutch sensor invalid fault ‘and ‘EPB acoustic warning invalid’	Liu Feng
2.6		4.4	Add the ‘note’ after table 4-6	Liu Feng
		5.1	Add the fault : 1. Auto Hold Switch Stuck ON : DTC : 0x51 0x63 0x00 2. Auto Hold Switch Invalid : DTC : 0x51 0x63 0x00 3. SSM_NO_FEEDBACK_PB_APPLIED:0x51 0x64 0x00 4. Engine Maximum Torque Invalid : DTC : 0x51 0x01 0x81 5. Engine Minimum Torque Invalid : DTC : 0x51 0x01 0x81 6. Engine Torque Constant Invalid : DTC : 0x51 0x01 0x81	
2.7	2012-3-7	4.4.1	Replace the C201 configuration example with CD101 implementation in point	Liu Feng
		4.4.2	Remove the variant fault of TCU detection	
		5.1	Add the warning lamp information for DTC	

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1. Introduction

1.1. Purpose

This document details the ECU addressing, supported diagnostic services, diagnostic routines, diagnostic identifiers, car configuration data and DTCs that are specific to the TRW CHANA CD101 ABS/VSC.

1.2. Symbol and abbreviation

A_PDU	Application layer_Protocol Data Unit，
BS	Block Size，
Client	Client side
CF	Consecutive Frame，
Cvt	Convention，
DATA	CAN data frame and data field
DLC	Data Length Code of the data frame
DSC	Diagnostic Session Control(SID = 0x10)
DTC	Diagnostic Trouble Code，
ECU	Electronic Control Unit，
FC	Flow Control，
FF	First Frame，
FF_DL	First Frame Data Length，
FS	Flow Status，
IDENTIFIER	CAN Identifier of the data frame
L_Data.req	L_Data.request，request data transmitted by Network layer to Data Link layer
L_Data.con	L_Data.confirm，confirmation data transmitted by Data Link layer to Network layer
L_Data.ind	L_Data.indication，indication data transmitted by Data Link layer to Network layer
L_PDU	Link data layer_Protocol Data Unit，
Lsb	least significant bit，
LSB	Least Significant Byte，
PCI	Protocol Control Information，
PDU	Protocol Data Unit，
msb	most significant bit，

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MSB	Most Significant Byte，
NRC	Negative Response Code，
NWL	NetWork Laye，
N_AE	Network layer Address Extension，
N_AI	Network layer Address Information，
N_Ar	Network layer timing parameter Ar，
N_As	Network layer timing parameter As，
N_Br	Network layer timing parameter Br，
N_Bs	Network layer timing parameter Bs，
N_Cr	Network layer timing parameter Cr，
N_Cs	Network layer timing parameter Cs，
N_Data	Network layer Data，
N_PCI	Network layer Protocol Control Information，
N_PCItype	Network layer Protocol Control Information type，
N_PDU	Network layer Protocol Data Unit，
N_SA	Network layer Source Address，
N_SDU	Network layer Service Data Unit，
N_TA	Netwok layer Target Address，
N_TAType	Netwok layer Target Address Type: Physical addressing and functional addrssing
N_USData	Netwok layer Unacknowledged Segmented Data， which is the interface service name between Network layer and Application layer
N_USData.req	N_USData.request， request data sent by Application layer to Network layer
N_USData.ind	N_USData.indication， indication data sent by Network layer to Application layer
N_USData.con	N_USData.confirm， confirmation data send by Network layer to Application layer
N_USData_FF	Unacknowledged Segmented Data First Frame，
N_USData_FF.ind	N_USData_FF.indication， indication data of the first frame of the multi-frames sent by Network layer to Application layer
OSI	Open System Interconnect，
PCI	Protocol Control Information，
SAE	Society of American Engineering，

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SF	Single Frame，
SF_DL	Single Frame Data Length，
SN	Sequence Number，
STmin	SeparationTime min.，
Server	Server: ECU
SID	Service Identifier，
SPRMIB	Suppress Positive Response Message Indication Bit，
USDT	Unacknowledged Segmented Data Transfer，
UUDT	Unacknowledged Un-segmented Data Transfer，

1.3. Related documents

1) ISO 15765-1 Road vehicles - Diagnostics on Controller Area Networks (CAN) - Part 1: General information
2) ISO 15765-2 Road vehicles - Diagnostics on Controller Area Networks (CAN) - Part 2: Network layer services
3) ISO 15765-3 Road vehicles - Diagnostics on Controller Area Networks (CAN) - Part 3: Implementation of unified diagnostic services (UDS on CAN)
4) ISO 14229-1 Road vehicles - Unified diagnostic services (UDS) - Part 1: Specification and requirements
5) ISO 15031-6 Road vehicles - Communication between vehicle and external equipment for emission-related diagnostics - Part 6: Diagnostic trouble code definitions
6) CHANA CD101 UDS on CAN V0.1 Initial

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2. Overview and protocol requirement from CHANA for CD101 Project -

2.1. Terms and definitions

The following terms and definitions defined in ISO 15765-1~3、ISO 14229-1、ISO 15031-6 are applicable to this section.

Server

Electronic Control Unit (ECU)，which responses to the diagnostic request.

Client

Diagnostic equipment, which sends the diagnostic request to the server.

1) Online diagnostic equipment in the vehicle: Diagnostic equipment fixed in the vehicle;
2) Offline diagnostic equipment: Connect to the network in the vehicle by diagnostic port.

Physical communication (addressing)

The diagnostic service between the client and the server(point to point).

Functional communication (addressing)

The client sends the same functional diagnostic request to multi-servers.

The diagnostic equipment can send the request message to ECU with any addressing mode, but ECU should send the response message to the diagnostic equipment with physical addressing.

Security status

Two state: Locked status and Un-locked status.

When ECU is powered on, the system will enter the security locked status of the restricted diagnostic service automatically.

ECU needs to enter the security access by Security Access service (SID = 0x27) to pass the security key check. After passing the security key check, ECU's security status will turn into Un-locked status, in which the diagnostic equipment will be able to request the restricted services.

Diagnostic conversation mode

Diagnostic conversation mode includes two types:

1) Default conversation mode;
2) No-default conversation mode.

After ECU is powered on, it will enter the default diagnostic conversation mode, in which the restricted diagnostic services are not supported. The Non-default diagnostic conversation mode can support the restricted diagnostic services. After ECU enters the Non-default diagnostic conversation mode, the diagnostic tool should maintain the diagnostic mode with Test Present service to avoid the timeout of S3 Timer; but when ECU is in the default diagnostic conversation mode, it doesn't need to maintain the diagnostic mode and timeout handling.

In this specification, two Non-default diagnostic modes are defined:

1) Programming mode;

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2) Extended mode.

Every diagnostic mode supports a set of special diagnostic services.

Parameter convention (Cvt)

Parameter convention requirement defines 4 types of the parameter usage:

1) M：Mandatory, which means the parameter must be used;
2) C：which means the parameter is used under some conditions;
3) S：Selective parameter, which means the parameter is mandatory and selected from the specified list;
4) U：User optional

Sub-functions

The different value indicates a different diagnostic service function.

Negative response codes

Negative response codes indicate the reasons why ECU can not send out the positive response.

Protocol data unit

Protocol data unit is a set of the information and data, which indicates the transmission information and data between the sender and the receiver. Protocol data unit includes:

1) Identification and addressing information;
2) Protocol control information(PCI)；
3) Data

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2.2. Network overview

2.2.1. Vehicle network

2.2.1.1. HS CAN network topology structure in the vehicle

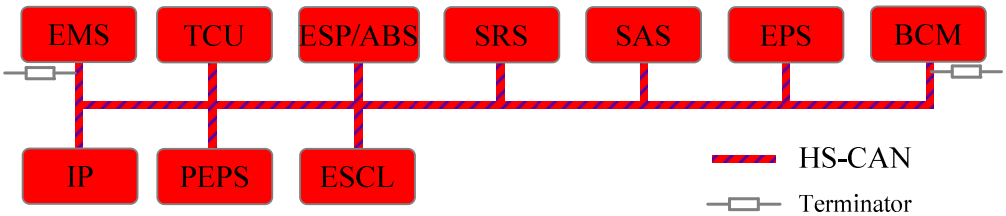

Diagram 1 CD101 HS CAN topology structure in the vehicle

2.2.1.2. Addressing mode

ECU's diagnostic function should use conventional addressing mode, not support the extended addressing mode and hybrid addressing mode.

2.3. Network layer parameters

2.3.1. Network layer concept

Network layer defines:

1) Network layer protocol data unit;
2) Network layer timing.

2.3.2. Interface service between Network layer and Application layer

The interface service between Network layer and Application layer includes:

1) Unacknowledged Segmented Data Transfer (USDT) service(N_USData)，which includes:
   - Request: Application layer requests Network layer to send the messages ;
   - Indication: Network layer reports Application layer having received the messages;
   - Confirmation: Network layer confirms the result of sending the messages to Application layer.
2) The first frame service of Unacknowledged Segmented Data Transfer (USDT) (N_USData_FF)，which includes:
   - Indication: Network layer indicates having received the first frame of the multi- frames data to Application layer.
3) Change Network layer parameters service (N_ChangeParameter)，which includes:
   - Request: Application layer requests Network layer to change Network layer parameters;
   - Confirmation: Network confirms the result of changing Network layer parameters to Application layer.

2.3.3. Network layer protocol data unit(N_PDU)

Table 1 N_PDU description

Parameter name	Abbreviation	Description

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Parameter name	Abbreviation	Description
Addressing information	N_AI	Include the information of Resource address, Target address, Addressing type and so on.
Protocol control information	N_PCI	Indicate N_PDU types: Single Frame(SF)、First Frame(FF)、Consecutive Frame(CF) and Flow Control Frame(FC)
Data	N_Data	Includes A_PCI and A_Data of Application layer

2.3.4. N_PCI of Single Frame N_PDU

N_PCI OF Single Frame N_PDU see Table 2，the data of the Single Frame Data Length (SF_DL) see Table 3.

Table 2 N_PCI of Single Frame N_PDU

N_PDU type	**N_PCI**		**Description**
	Byte0		
	Bit7~4	**Bit3~0**	
Single Frame(SF)	0	SF_DL	Indicates Single Frame

Table 3 SF_DL Data Definition

Data	Description
0	reserved
1~7	The data length of the Single Frame, unit: byte
8~15	invalid

If SF_DL = reserved or invalid then the receiving side will ignore the received Single Frame and its Network layer will not use N_USData.ind to report to Application layer.

2.3.5. N_PCI of the First Frame N_PDU

For N-PCI_of the First Frame N_PDU, see Table 4，for the data length (FF_DL)definition of the multi-frames, see Table 5.

Table 4 N_PCI of the First Frame N_PDU

N_PDU tyoe	**N_PCI**			**Description**
	Byte0		**Byte1**	
	Bit7~4	**Bit3~0**		
First Frame(FF)	1	FF_DL		Indicates the First Frame of the multi-frames

Table 5 FF_DL data definition

Data	Description
0~7	invalid

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Data	Description
8~4095	Data length of multi-frames, unit: byte

If FF_DL = invalid，then the receiving side will stop receiving the messages, will not send the Flow Control Frame, and its Network layer will not use N_USData.ind to report to Application layer.

If FF_DL is more than the buffer size of the receiving side, then the receiving side will stop receiving the messages, will send Flow Control Frame with FS = Overflow(at this time the BS and STmin of the Flow Control Frame make no sense), and its Network layer will not use N_USData_FF.ind to report to Application layer. After the Network layer of the sending side receives this Flow Control Frame, it should use N_USData.con to report N_Result = N_Buffer_Overflow to Application layer. The Application layer of the sending side should not start the timer of the Application layer after receiving this N_Result.

2.3.6. N_PCI of the Consecutive Frames N_PDU

For N_PCI of the Consecutive Frames N_PDU, see Table 6; the Serial Number (SN) data definition of the Consecutive Frames, see Table 7.

Table 6 N_PCI of Consecutive Frames N_PDU

N_PDU type	**N_PCI**		**Description**
	Byte0		
	Bit7~4	**Bit3~0**	
Consecutive Frames(CF)	2	SN	Indicates the Consecutive Frames of the multi- frames

Table 7 SN data definition

Data	Description
0~15	Indicates the Serial Number of the Consecutive Frames, the Serial Number of the first Consecutive Frame is 1, its value is increased by 1. If the Serial Number is more than 15, it will be started from 0 again.

If SN doesn't accord with the counting ruler, the receiving side will stop receiving the messages and handling them accordingly, meanwhile the Network layer of the receiving side will use N_USData.ind to report N_Result = N_Wrong_SN to Application layer.

2.3.7. N_PCI of the Flow Control Frame N_PDU

For N_PCI of the Flow Control Frame N_PDU see Table 8; for the data definition of Flow Control Status(FS)、continuously Sending times (BS)、Smallest time interval(STmin) see Table 9~11.

Table 8 N_PCI of Flow Control Frame N_PDU

N_PDU type	**N_PCI**				**Description**
	Byte0		**Byte1**	**Byte2**	
	Bit7~4	**Bit3~0**			

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N_PDU type	N_PCI				Description
	Byte0		Byte1	Byte2	
	Bit7~4	Bit3~0			
Flow Control Frame(FC)	3	FS	BS	STmin	Used for the sending control of the multi-frames by the receiving side.

Table 9 FS data definition

Data	Description
0	ContinueToSend indicates the sending side is allowed to continue sending the Consecutive messages.
1	Wait indicates the sending side needs to wait for the next Flow Control Frame[1],which is called Wait Flow Control Frame.
2	Overflow indicates that the length of the messages exceeds the buffer size of the Network layer of the receiving side. This Flow Control Frame will force the sending side to stop sending the multi-frames, and the Network layer of the sending side will use N_USData.con to report N_Result = N_Buffer_Overflow to Application layer. The Flow Control Frame of FS = Overflow will be sent out by the receiving side only after the first frame is received.
3~15	reserved
Note:	[1] The continuously sending times of the Flow Control Frame by the sending side can not be more than the Maximal N_WFT value, the parameter N_WFT value is decided during the system design.

Table 10 BS data definition

Data	Description
0	Indicates that the sending side is allowed to send the Consecutive Frames and doesn't need to wait for the Flow Control Frame sent by the receiving side.
1~255	Indicates the allowed sending times (Number) of the Consecutive Frames continuously. After finishing sending the specified Consecutive Frames, the sending side must wait for the Flow Control Frame from the receiving side.

Table 11 STmin data definition

Data	Description
0~127	The time interval between two Consecutive Frames, unit: ms
128~255	reserved

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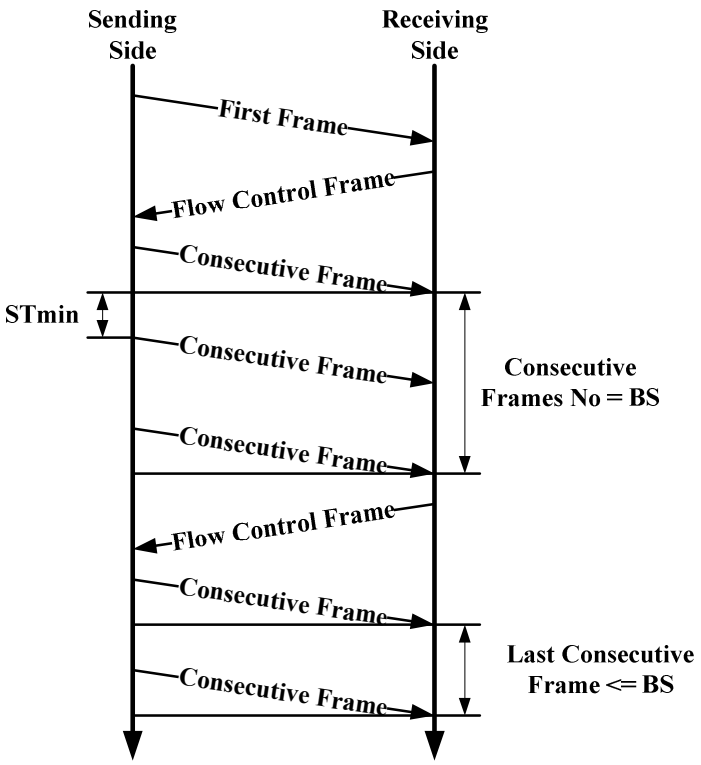

Diagram 2 BS and STmin diagram (BS=3)

2.3.8. N_PDU and L_PDU

Table 12 N_PDU and L_PDU

N_PDU type	L_PDU						
	IDENTIFIER		DLC	DATA[1]			
	Bit10~8	Bit7~0		Byte0	Byte1	Byte2	Byte3~7
Single Frame(SF)	7	N_AI	8	N_PCI	N_Data		
First Frame(FF)	7	N_AI	8	N_PCI		N_Data	
Consecutive Frame(CF)	7	N_AI	8	N_PCI	N_Data		
Flow Control(FC)	7	N_AI	8	N_PCI			0x00

Note: [1] As per filling ruler, if the valid data length of the Single Frame and the last Consecutive Frame is less than 8 bytes, the unused bytes will be padded with 0x00; If the valid data of Flow Control Frame are 3 bytes, the last 5 bytes should be padded with 0x00.

2.3.9. Network layer timing

2.3.9.1. Timing description

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For Network layer timing see Table 13, they are using the interface service of the Data Link layer and Network layer as the start and stopping time point. For details, see Table 14-15.

Table 13 Network layer timing

Timing	Description
N_As/N_Ar	Sending time of CAN data frame through Data Link layer
N_Bs	Waiting time for receiving Flow Control Frame of the sending side
N_Br	Interval time for sending the Flow Control Frame of the receiving side
N_Cs	Interval time for sending the Consecutive Frames of the sending side
N_Cr	Waiting time for receiving the Consecutive Frames of the receiving side

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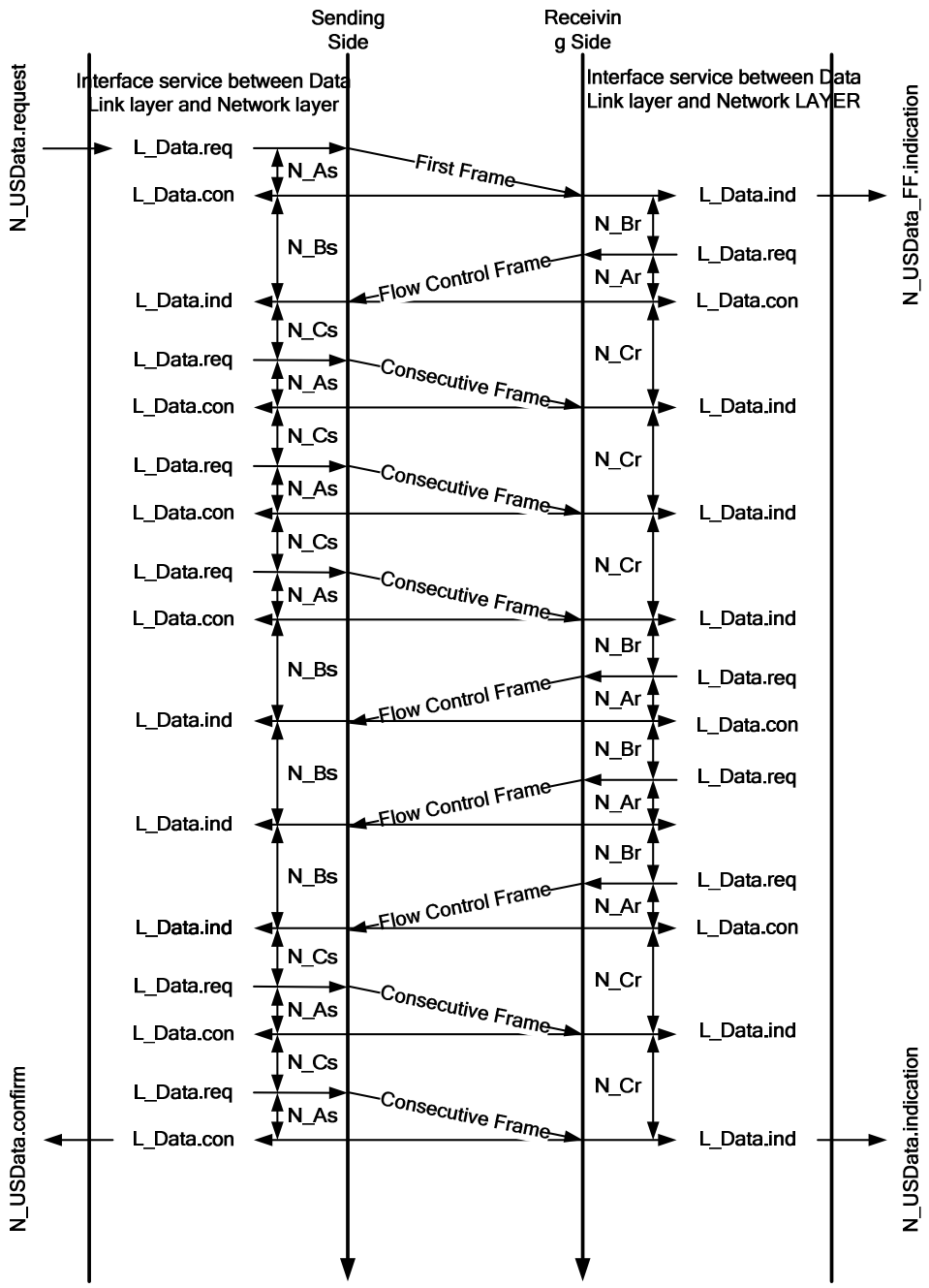

Diagram 3 Network layer timing

2.3.9.2. Network layer parameters requirement

Table 14 Network layer parameters

Parameter	Symbol	Application mode[1]

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Parameter	Symbol	Application mode[1]
Maximal sending times (Number) of Waiting Flow Control Frame	N_WFT_{MAX}	0
Continuously sending times (Number)	BS	8
Smallest time interval	STmin	20

Table 15 Network layer timing

Timing	Overtime timing[1] (ms)	Performance requirement(ms)
N_As/N_Ar	70	-
N_Bs	150	-
N_Br	-	<70
N_Cs	-	<70
N_Cr	150	-

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2.4. Application layer parameters

Application layer concept

The following contents are described in Application layer:

1) Application layer Protocol Data Unit(A_PDU);
2) Application layer conversation management timing.

Application layer Protocol Data Unit(A_PDU)

Table 16 A_PDU Description

Parameter Name	Abbreviation	Description
Addressing Information	A_AI	Includes Source Address, Target Address and Addressing modes and so on
Protocol Control Information	A_PCI	Includes diagnostic service Identifier(SID/RSID)
Data	A_Data	Includes the sub-functions of the diagnostic services and their data contents

A_PDU of diagnostic service request data

The data length of Diagnostic service request data A_PCI is 1 byte，which is used for storing SID.

Table 17 Sub-function byte

Bit7	Bit6~0
SPRMIB，Inhibit Positive Response Indicating Bit[1]	Sub-function value
Note: [1] The setting of Inhibit Positive Response Indicating Bit doesn't effect the negative response.	

Table 18 Data definition of Inhibit Positive Response Indicating Bit (SPRMIB)

Data	Description
0	FALSE，doesn't inhibit the server's positive response
1	TRUE，does inhibit the server's positive response

A_PDU of diagnostic service response data

The data length of diagnostic service positive response A_PCI is 1 byte, which is used for storing RSID (RSID= Corresponding request data SID+0x40).

The data length of A_Data，which is used for storing NRC.

Application layer conversation management timing

Table 19 Application layer timing

Timing	Description
$P2_{CAN_Server}$	Timing requirement for the server to send out the response data after receiving the request data
$P2_{CAN_Client}$	The overtime setting for the Client to wait for the response data from the server after the request data are transmitted successfully(FF or SF)

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Timing	Description
$P2^*_{CAN_Server}$	Timing requirement for the server to send out the response data after the server sends out the negative response (negative response code 0x78)
$P2^*_{CAN_Client}$	The enhanced overtime timing setting for the client to wait for the response data from the server after the client receives the negative response (negative response code 0x78) (FF or SF)
$P3_{CAN_Client_Phy}$	The smallest time interval from the successfully sending the physical addressing request of the client (this request doesn't need the server to response) to the next sending the physical addressing request
$P3_{CAN_Client_Func}$	The smallest time interval from the successfully sending the functional addressing request of the client to the next sending the functional addressing request, which includes two cases: 1. the request from the client doesn't need to get the response from the server; 2. the request will only be replied by some of the servers.
$S3_{Server}$	Time parameter of the server, which is only used for non-default conversation mode. During the period of $S3_{Server,}$ if the server doesn't receive any diagnostic request data then it will exit the non-default conversation mode and return to the default conversation mode.
$S3_{Client}$	Timing parameter of the client: the time interval between the two consecutive TestPresent request messages for the client to keep the connection of the non-default conversation mode.

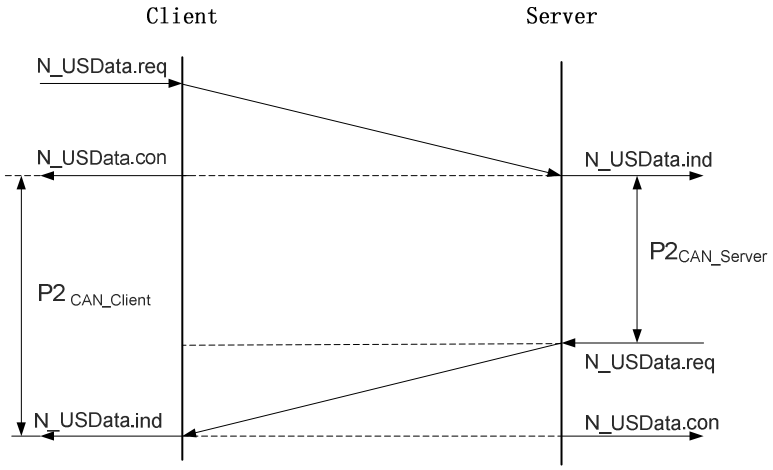

Diagram 1 Application layer timing parameters

Application layer conversation management timing

Table 20 Application layer timing

Timing	Client	Server	Min(ms)	Nominated(ms)	Max(ms)

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Timing	Client	Server	Min(ms)	Nominated(ms)	Max(ms)
$P2_{CAN_Server}$	-	√	0	-	50
$P2_{CAN_Client}$	√	-	100	-	-
$P2^*_{CAN_Server}$	-	√	0	-	5000
$P2^*_{CAN_Client}$	√	-	5050	-	-
$P3_{CAN_Client_Phy}$	√	-	100	-	-
$P3_{CAN_Client_Func}$	√	-	100	-	-

Table 6 S3 Conversation management timing

Timing	Client	Server	Min(ms)	Nominated(ms)	Max(ms)
$S3_{Server}$	-	√	-	5000	-
$S3_{Client}$	√	-	-	4000	-

Timing fault handling

Table 7 Fault handling of the client

Communication stage	Client fault type	Fault handling of the client[1]	
		Physical communication	Functional communication
Sending request	N_USData.con feedbacks the sending fault of the client	After $P3_{CAN_Client_Phy}$ is overtime, the client will re-send a request message, and will restart the timing parameter $S3_{Client}$ in non-default mode.	After $P3_{CAN_Client_Fun}$ is overtime, the client will re-send the last request message.
$P2_{CAN_Client}$ $P2^*_{CAN_Client}$	overtime	The client will re-send the last request message, and will re-start the timing parameter $S3_{Client.}$	If the client does not know the response node, the fault will not be handled after the overtime. If the client knows the response node, the client will re-start the last request message after the overtime.

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Communication stage	Client fault type	Fault handling of the client[1]	
		Physical communication	Functional communication
Receiving response	N_USData.ind reports the receiving fault of the client	The client will re-send the last request message, and will re-start the timing parameter $S3_{Client}$ in the non-default mode.	The client will re-send the last request message.
Note:	[1] The client is allowed to handle the fault at most twice.		

Table 8 Fault handling of the server

Communiocation stage	Server fauld type	Fault handling of the server
Receiving request	N_USData.ind reports the receiving fault of the server	Ignore the received diagnostic service request, re-start the timing parameter $S3_{Server}$
$P2_{CAN_Server}$ $P2^*_{CAN_Server}$	overtime	-
Sending response	N_USData.con feedbacks the sending fault of the server	Does not send the response message any more, re-start the timing parameter $S3_{Server}$

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3. Supported diagnostic services in details for CHANA CD101 ABS/ESP

3.1. ECU Addressing

The ECU Addresses supported by the ECU are:

CAN ID	Description
780	Physically Addressed Diagnostic Request
788	Physically Address ECU Response
7DF	Functional Addressed Request

Table 3-1: ECU Addresses

The ECU must respond with a physically addressed response regardless of if the request was functionally or physically addressed.

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3.2. Supported Diagnostic Services overview

Note: The detailed diagnostic services are specified in CHANA Diagnostic Services Specification.

Service name	SID	Default Session		Extended Session		Sub-functions	Supported Conditions	Comments
		Phys	Func	Phys	Func			
DiagnosticSessionControl	10	✓	✓	✓	✓	Default Session		
						Default Session (0x01)		
						Programming Session (0x02)		
						Extended Session (0x03)		
						EOL (End of Line Session) (0x04)		
EcuReset	11	✓	✓	✓	✓	hardReset (0x01)		
SecurityAccess	27			✓		Manufacturer		
						ReProgramming		
						System Supplier		
CommunicationControl	28			✓	✓	enableRxAndTx (0x00)		
						disableRxAndTx(0x03)		
TesterPresent	3E	✓	✓	✓	✓	zeroSubFunction (0x00)		
ControlDTCSetting	85			✓	✓	On (0x01)		
						Off (0x02)		
ReadDataByIdentifier	22	✓	✓	✓	✓			
ReadMemoryByAddress	23							
WriteDataByIdentifier	2E			✓				

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Service name	SID	Default Session		Extended Session		Sub-functions	Supported Conditions	Comments
		Phys	Func	Phys	Func			
WriteMemoryByAddress	3D							
ClearDiagnosticInformation	14	✓	✓	✓	✓	All Groups (FFFFFF)		
ReadDTCInformation	19	✓	✓	✓	✓	reportNumberOfDTCByStatusMask		
						reportDTCByStatusMask		
						reportDTCSnapshotIdentification		
						reportDTCSnapshotRecord-ByDTCNumber		
						reportDTCExtendedDataRecord-ByDTCNumber		
						reportSupportedDTC		
InputOutputControlByIdentifier	2F			✓		returnControlToECU		
						shortTermAdjustment		
RoutineControl	31			✓				See Routine Sheet
RequestDownload	34							Only in Programming session
TransferData	36							Only in Programming session
RequestTransferExit	37							Only in Programming session

Table 3-2: Supported Diagnostic Services

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3.3. Diagnostic services

3.3.1. DiagnosticSessionControl (DSC) (SID = 0x10)

The ECU shall switch to the requested non-default session only if the following conditions are satisfied:

1) Vehicle speed < 3 kmph;
2) No Control Algorithm is Active.

3.3.1.1. Service Request

Table 1 DSC Service Request

A_PDU Parameter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI	DiagnosticSessionControl Req ID	M	0x10	DSC
A_Data[0]	sub-function = [diagnosticSessionType]	M	0x00~0x7F	LEV_ DS_

Table 2 DiagnosticSessionType(LEV_DS_) data definition

Data(Hex)	Description	Cvt	Abbreviation
0x00	reserved	M	-
0x01	defaultSession，	M	DS
0x02	programmingSession，	M	PRGS
0x03	extendedDiagnosticSession，	M	EXTDS
others	reserved	U	-

3.3.1.2. Response messages

Table 3 DSC Positive response

A_PDU Paramter	Paramter name	Cvt	Data(Hex)	Abbreviation
A_PCI	DiagnosticSessionControl Resp Id	S	0x50	DSCPR
A_Data[0]	diagnosticSessionType	M	0x00~0x7F	DS_
	sessionParameterRecord[] = [			SPREC_
A_Data[1]	$P2_{CAN_Server_max}$ (MSByte)	M	0x00~0xFF	P2CSMH
A_Data[2]	$P2_{CAN_Server_max}$ (LSByte)	M	0x00~0xFF	P2CSML
A_Data[3]	$P2^*_{CAN_Server_max}$ (MSByte)	M	0x00~0xFF	P2ECSMH
A_Data[4]	$P2^*_{CAN_Server_max}$ (LSByte)]	M	0x00~0xFF	P2ECSML

Table 4 DSC Negative response

A_PDU Parameter	Paramter name	Cvt	Data(Hex)	Abbreviation

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A_PDU Parameter	Paramter name	Cvt	Data(Hex)	Abbreviation
A_PCI[0]	negativeResponseSID	M	0x7F	SIDNR
A_PCI[1]	DiagnosticSessionControl Req ID	M	0x10	DSC
A_Data[0]	negativeResponseCode	M	0x12, 0x13, 0x22	NRC_

Table 5 DSC Negative response codes (NRC_)

Data(Hex)	Description	Cvt	Abbreviation
0x12	subFunctionNotSupported	M	SFNS
0x13	incorrectMessageLengthOrInvaliformat	M	IMLOIF
0x22	conditionsNotCorrect	M	CNC

3.3.1.3. Diagnostic sessions state transferring: See Diagram 1

Note: 1) After startup the ECU is in default diagnostic session mode (10 01).

2) Timeout from Non-default session to Default diagnostic session (5 seconds).

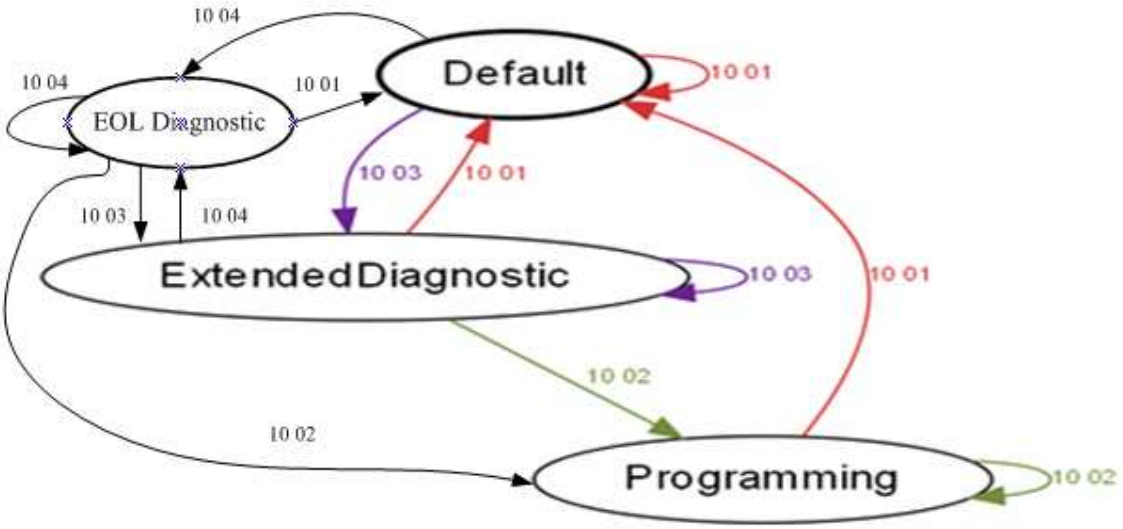

Diagram 1 Diagnostic sessions state transferring

Note: The EOL tests for TRW ABS/VSC at High speeds can only be conducted in EOL diagnostic session (10 04).

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3.3.2. ECUReset(ECUR) (SID = 0x11)

This service shall be available only if the following conditions are satisfied:

1) Vehicle speed < 3kmph
2) No Control Algorithm is Active
3) At least one Wheel Speed Sensor is valid

3.3.2.1. Service Request

Table 1 ECUR Service request

A_PDU Paramter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI	ECUReset Req Id	M	0x11	ER
A_Data[0]	sub-function=[resetType]	M	0x00~0x7F	LEV_ RT_

Table 2 ResetType(LEV_RT_) Data definition

Data(Hex)	Description	Cvt	Abbreviation
0x00	reserved	M	-
0x01	hardReset，	U	HR
0x02	keyOffOnReset，	U	KOFFONR
0x03	softReset，	U	SR
0x04~0x7F	reserved	M	-

3.3.2.2. Response messages

Table 1 ECUR Positive response

A_PDU Paramter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI	ECUReset Resp ID	S	0x51	ERPR
A_Data[0]	resetType	M	0x00~0x7F	RT_

Table 2 ECUR Negative response

A_PDU Paramter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI[0]	negativeResponseSID	M	0x7F	SIDNR
A_PCI[1]	ECUReset Req Id	M	0x11	ER

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A_PDU Paramter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_Data[0]	negativeResponseCode	M	0x12, 0x13, 0x22	NRC_

Table 3 ECUR Negative response codes (NRC_)

Data(Hex)	Description	Cvt	Abbreviation
0x12	subFunctionNotSupported	M	SFNS
0x13	incorrectMessageLengthOrInvaliormat	M	IMLOIF
0x22	conditionsNotCorrect	M	CNC
33	SecurityAccessDenied	M	SAD

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3.3.3. SecurityAccess (SA) (SID = 0x27)

Check CHANA Logic Algorithm specification for Security Access Level 1.
The other Security Level (for example Programming Security Level or Supplier Security Level) and its logic algorithm can be defined by TRW.

3.3.3.1. Service Request

Table 1 SA Service request (sub-function = requestSeed)

A_PDU Parameter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI	SecurityAccess Req ID	M	0x27	SA
A_Data[0]	sub-function = [securityAccessType = requestSeed]	M	0x01	LEV_ SAT_RSD

Table 2 SA Request message (sub-function = sendKey)

A_PDU Paramter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI[0]	requestSID	M	0x27	SIDRQ
A_Data[0]	sub-function = [securityAccessType = sendKey]	M	0x02	LEV_ SAT_SK
A_Data[1] …… A_Data[m]	securityKey[][1] = [key#1 (MSByte) …… key#m (LSByte)]	M …… U	0x00~0xFF …… 0x00~0xFF	SECKEY_ KEY1HB …… KEYmLB
Note	[1] The data length of Securitykey is decided by the Security logic。			

3.3.3.2. Response messages

Table 1 SA Positive response (sub-function = requestSeed)

A_PDU Paramter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI[0]	SecurityAccess Resp Id	S	0x67	SAPR
A_Data[0]	securityAccessType = requestSeed	M	0x01	SAT_RSD
A_Data[1] …… A_Data[m]	securitySeed[] = [seed#1 (MSByte) ..... seed#m (LSByte)]	M …… U	0x00~0xFF …… 0x00~0xFF	SECSEED_ SEED1HB …… SEEDmLB

Table 2 SA Positive response (sub-function = sendKey)

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A_PDU Paramter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI	SecurityAccess Resp Id	S	0x67	SAPR
A_Data[0]	securityAccessType = sendKey	M	0x02	SAT_SK

Table 3 SA Negative response

A_PDU Paramter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI[0]	negativeResponseSID	M	0x7F	SIDNR
A_PCI[1]	SecurityAccess Req ID	M	0x27	SA
A_Data[0]	negativeResponseCode	M	0x12, 0x13, 0x22, 0x24, 0x31, 0x35, 0x36, 0x37	NRC_

Table 4 SA Negative response codes (NRC_)

Data(Hex)	Description	Cvt	Abbreviation
0x12	subFunctionNotSupported	M	SFNS
0x13	incorrectMessageLengthOrInvalidformat	M	IMLOIF
0x22	conditionsNotCorrect	M	CNC
0x24	requestSequenceError	M	RSE
0x31	requestOutOfRange	M	ROOR
0x35	invalidKey	M	IK
0x36	exceededNumberOfAttempts	M	ENOA
0x37	requiredTimeDelayNotExpired [1]	M	RTDNE
Note	[1] If 3 attempts of entering security access fail, the server will not be allowed to do the security access attempt within 10 seconds.		

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3.3.4. CommunicationControl (CC) (SID = 0x28)

3.3.4.1. Service Request

Table 1 CC Service request

A_PDU Paramter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI	CommunicationControl Req ID	M	0x28	CC
A_Data[0]	sub-function = [controlType]	M	0x00~0x7F	LEV_ CTRLTP_
A_Data[1]	communicationType	M	0x00, 0x01, 0x02, 0x03	CTP

Table 2 controlType(LEV_CTRLTP_) Data definition

Data(Hex)	Descriptioon	Cvt	Abbreviation
0x00	enableRxAndTx，	M	ERXTX
0x01	enableRxAndDisableTx，	M	ERXDTX
0x02	disableRxAndEnableTx，	M	DRXETX
0x03	disableRxAndTx，	M	DRXTX
0x04~0x7F	reserved	M	-

Table 3 communicationType(CTP) Data definition

Data(Hex)	Description	Cvt	Abbreviation
0x01	Normal communication messages	M	NCM
0x02	Network management messages	M	NWMCM
0x03	Normal and network communication messages	M	NCM_ NWMCM

3.3.4.2. Response messages

Table 1 CC Positive response

A_PDU Parameter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI	CommunicationControl Resp ID	S	0x68	CCPR
A_Data[0]	controlType	M	0x00~0xFF	CTRLTP

Table 2 CC Negative response

A_PDU Parameter	Paramter Name	Cvt	Data(Hex)	Abbreviation
A_PCI[0]	negativeResponseSID	M	0x7F	SIDNR

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A_PDU Parameter	Paramter Name	Cvt	Data(Hex)	Abbreviation
A_PCI[1]	CommunicationControl Req ID	M	0x28	CC
A_Data[0]	negativeResponseCode	M	0x12, 0x13, 0x22, 0x31	NRC_

Table 3 CC Negative response codes (NRC_)

Data(Hex)	Description	Cvt	Abbreviation
0x12	subFunctionNotSupported	M	SFNS
0x13	incorrectMessageLengthOrInvalidFormat	M	IMLOIF
0x22	conditionsNotCorrect	M	CNC
0x31	requestOutOfRange	M	ROOR

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3.3.5. TesterPresent(TP) (SID = 0x3E)

CHANA Suggestion: For functional addressing TP, SPRMIB = True; for physical addressing TP, SPRMIB = False.

3.3.5.1. Service Request

Table 1 TP Service request

A_PDU Paramter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI	TesterPresent Req ID	M	0x3E	TP
A_Data[0]	sub-function = [zeroSubFunction]	M	0x00	LEV_ ZSUBF

3.3.5.2. Response messages

Table 2 TP Positive response

A_PDU Paramter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI	TesterPresent Resp ID	S	0x7E	TPPR
A_Data[0]	zeroSubFunction	M	0x00	ZSUBF

Table 3 TP Negative response

A_PDU Parameter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI[0]	negativeResponseSID	M	0x7F	SIDNR
A_PCI[1]	TesterPresent Req ID	M	0x3E	TP
A_Data[0]	negativeResponseCode	M	0x12, 0x13	NRC_

Table 4 TP Negative response codes (NRC_)

Data(Hex)	Description	Cvt	Abbreviation
0x12	subFunctionNotSupported	M	SFNS
0x13	incorrectMessageLengthOrInvalidFormat	M	IMLOIF

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3.3.6. ControlDTCSetting (CDTCS) (SID = 0x85)

Entry conditions for service $85:

1) v < 3 km/h
2) No control algorithm is active
3) At least one Wheel Speed Sensor is valid

3.3.6.1. Service Request

Table 1 CDTCS Service request

A_PDU Paramter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI	ControlDTCSetting Req ID	M	0x85	CDTCS
A_Data[0]	sub-function = [DTCSettingType]	M	0x00~00xFF	LEV_ DTCSTP_
A_Data[1] …… A_Data[n]	DTCSettingControlOptionRecord[]=[Parameter#1 …… Parameter#1]	U …… U	0x00~00xFF …… 0x00~00xFF	DTCSCOR_ PARA1 …… PARAm

Table 2 DTCSettingType(LEV_DTCSTP_) Data definition

Data(Hex)	Description	Cvt	Abbreviation
0x00	ISO/SAE reserved	M	ISOSAERESRVD
0x01	On Switch on DTC setting	U	ON
0x02	Off Switch off DTC setting	U	OFF
0x03~0x3F	ISO/SAE reserved	U	ISOSAERESRVD
0x40~0x5F	Vehicle manufacturer define	U	VMS
0x60~0x7E	System supplier define	U	SSS
0x7F	ISO/SAE reserved	M	ISOSAERESRVD

3.3.6.2. Response Messages

Table 1 CDTCS Positive response

A_PDU Parameter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI	ControlDTCSetting Resp ID	S	0xC5	CDTCSPR
A_Data[0]	DTCSettingType	M	0x00~00xFF	DTCSTP

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Table 2 CDTCS Negative response

A_PDU Parameter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI[0]	negativeResponseSID	M	0x7F	SIDNR
A_PCI[1]	ControlDTCSetting Req ID	M	0x85	DTCSTP
A_Data[0]	negativeResponseCode	M	0x12, 0x13, 0x22, 0x31	NRC_

Table 3 CDTCS Negative response codes (NRC_)

Data(Hex)	Description	Cvt	Abbreviation
0x12	subFunctionNotSupported	M	SFNS
0x13	incorrectMessageLengthOrInvalidFormat	M	IMLOIF
0x22	conditionsNotCorrect	M	CNC
0x31	requestOutOfRange	M	ROOR

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3.3.7. ReadDataByIdentifier (RDBI) (SID = 0x22)

3.3.7.1. Service Request

Table 1 RDBI Service request

A_PDU Paramter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI	ReadDataByIdentifier Req ID	M	0x22	RDBI
A_Data[0] A_Data[1]	dataIdentifier[] #1 = [byte#1 (MSByte) byte#2 (LSByte)]	 M M	 0x00~0xFF 0x00~0xFF	DID_ B1 B2
……	……	……	……	……
A_Data[n-1] A_Data[n]	dataIdentifier[] #(n+1)/2 = [byte#1 (MSByte) byte#2 (LSByte)]	 U U	 0x00~0xFF 0x00~0xFF	DID_ B1 B2

Table 2 DataIdentifier(DID_) Data definition

3.3.7.2. Response messages

Table 1 RDBI Positive response

A_PDU Parameter	Parameter	Cvt	Data(Hex)	Abbreviation
A_PCI	ReadDataByIdentifier Resp ID	M	0x62	RDBIPR
A_Data[0] A_Data[1]	dataIdentifier[] #1 = [byte#1 (MSByte) byte#2 (LSByte)]	 M M	 0x00~0xFF 0x00~0xFF	DID_ B1 B2
A_Data[2] …… A_Data[k+1]	dataRecord[] #1 = [data#1 (MSByte) …… data#k (LSByte)]	 M …… U	 0x00~0xFF …… 0x00~0xFF	DREC_ DATA_1 …… DATA_k
……	……	……	……	……
……	dataIdentifie[] r#(n+1)/2 = [byte#1 (MSByte) byte#2 (LSByte)]	 M M	 0x00~0xFF 0x00~0xFF	DID_ B1 B2

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A_PDU Parameter	Parameter	Cvt	Data(Hex)	Abbreviation
……	dataRecord[] #(n+1)/2 = [data#1 (MSByte) …… data#k (LSByte)]	 M …… U	 0x00~0xFF …… 0x00~0xFF	DREC_ DATA_1 …… DATA_k

Table 2 RDBI Negative response

A_PDU Parameter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI[0]	negativeResponseSID	M	0x7F	SIDNR
A_PCI[1]	ReadDataByIdentifier Req ID	M	0x22	RDBI
A_Data[0]	negativeResponseCode	M	0x13, 0x22, 0x31	NRC_

Table 3 RDBI Negative response codes (NRC_)

Data(Hex)	Description	Cvt	Abbreviation
0x13	incorrectMessageLengthOrInvalidFormat	M	IMLOIF
0x22	conditionsNotCorrect	M	CNC
0x31	requestOutOfRange	M	ROOR

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3.3.8. WriteDataByIdentifier (WDBI) (SID = 0x2E)

3.3.8.1. Service Request

Table 1 WDBI Service request

A_PDU Paramter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI	WriteDataByIdentifier Req ID	M	0x2E	WDBI
	dataIdentifier[] = [			DID_
A_Data[0]	byte#1 (MSByte)	M	0x00~0xFF	B1
A_Data[1]	byte#2 (LSByte)]	M	0x00~0xFF	B2
	dataRecord[] = [			DREC_
A_Data[2]	data#1 (MSByte)	M	0x00~0xFF	DATA_1
.....	.....	.....	.....	.....
A_Data[n]	data#k (LSByte)]	U	0x00~0xFF	DATA_k

3.3.8.2. Response Messages

Table 2 WDBI Positive response

A_PDU Paramter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI	WriteDataByIdentifier Resp ID	M	0x6E	WDBIPR
	dataIdentifier[] = [			DID_
A_Data[0]	byte#1 (MSByte)	M	0x00~0xFF	B1
A_Data[1]	byte#2 (LSByte)]	M	0x00~0xFF	B2

Table 3 WDBI Negative response

A_PDU Parameter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI[0]	negativeResponseSID	M	0x7F	SIDNR
A_PCI[1]	WriteDataByIdentifier Req ID	M	0x2E	WDBI
A_Data[0]	negativeResponseCode	M	0x13, 0x22, 0x31, 0x33, 0x72	NRC_

Table 4 WDBI Negative response codes (NRC_)

Data(Hex)	Description	Cvt	Abbreviation
0x13	incorrectMessageLengthOrInvalidFormat	M	IMLOIF
0x22	conditionsNotCorrect	U	CNC

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Data(Hex)	Description	Cvt	Abbreviation
0x31	requestOutOfRange	M	ROOR
0x33	securityAccessDenied	M	SAD
0x72	generalProgrammingFailure	M	GPF

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3.3.9. ClearDiagnosticInformation (CDTCI) (SID = 0x14)

This service shall be available only if the following conditions are satisfied:

1) Vehicle speed < 3kmph
2) No Control Algorithm is Active.
3) At least one Wheel Speed Sensor is valid.

NOTE: In case of 4 invalid WSS, Clear DTC is possible if the unfiltered wheel velocities are below 3 km/h

3.3.9.1. Service Request

Table 1 CDTCI Service request

A_PDU Parameter	Parameter name	Cvt	Data(Hex	Abbreviation
A_PCI	ClearDiagnosticInformation Req ID	M	0x14	CDTCI
	groupOfDTC[][1] = [			GODTC_
A_Data[0]	DTC (HighByte)	M	0x00~0xFF	HB
A_Data[1]	DTC(MiddleByte)	M	0x00~0xFF	MB
A_Data[2]	DTC (LowByte)]	M	0x00~0xFF	LB
Note	[1] DTC group definition			

3.3.9.2. Response Messages

Table 2 CDTCI Positive response

A_PDU Parameter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI	ClearDiagnosticInformation Resp ID	M	0x54	CDTCIPR

Table 3 CDTCI Negative response

A_PDU Parameter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI[0]	negativeResponseSID	M	0x7F	SIDNR
A_PCI[1]	ClearDiagnosticInformation Req ID	M	0x14	CDTCI
A_Data[0]	negativeResponseCode	M	0x13, 0x22, 0x31	NRC_

Table 4 CDTCI Negative response codes (NRC_)

Data(Hex)	Description	Cvt	Abbreviation
0x13	incorrectMessageLengthOrInvalidFormat	M	IMLOIF
0x22	conditionNotCorrect	C	CNC

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Data(Hex)	Description	Cvt	Abbreviation
0x31	requestOutOfRequest	M	ROOR

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3.3.10. ReadDTCInformation (RDTCI) (SID = 0x19)

3.3.10.1. Sub-funtions overview:

This service is used to reporting the supported DTC information by the server.It includes the below sub-functions:

1) Report the number of the DTCs by status mask(0x19 0x01)；
2) Report all DTCs list by status mask (0x19 0x02)；
3) Report the DTCSnapshot information records regarding DTC and status mask (0x19 0x03、0x19 0x04)；
4) Report the extended data records of the DTC (0x19 0x06)；
5) Report all supported state of the DTC (0x19 0x0A)。

3.3.10.1.1. Report the number of the DTCs by status mask

This sub-function is used for reporting the DTC number by status mask.

3.3.10.1.2. Report all DTCs list by status mask

This sub-function is used for reporting all DTC list by status mask.

3.3.10.1.3. Report DTCSnapshot information records

This sub-function is used for reporting DTCSnapshot information records, the response messages include the DTCSnapshot records number and records information.

3.3.10.1.4. Report DTC extended data

This sub-function is used for reporting one or all extended data of the DTC.

3.3.10.1.5. Report all supported state of the DTC

This sub-function is used for reporting all supported state of the DTC. The response messages include the DTC valid status mask, DTC and DTC state records.

3.3.10.2. Service Request

Table 1 RDTCI Service request

(sub-function = reportNumberOfDTCByStatusMask，reportDTCByStatusMask)

A_PDU Parameter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI	ReadDTCInformation Req ID	M	0x19	RDTCI
A_Data[0]	sub-function = [reportNumberOfDTCByStatusMask	M	0x01	LEV_ RNODTCBSM
	reportDTCByStatusMask]	M	0x02	RDTCBSM
A_Data[1]	DTCStatusMask	M	0x00~0xFF	DTCSM

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Table 2 RDTCI Service request

(sub-function = reportDTCSnapshotIdentification，reportDTCSnapshotRecordByDTCNumber)

A_PDU Parameter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI	ReadDTCInformation Req ID	M	0x19	RDTCI
A_Data[0]	sub-function = [reportDTCSnapshotIdentification	M	0x03	LEV_ RNODTCBSM
	reportDTCSnapshotRecordByDTC Number]	M	0x04	RDTCBSM
A_Data[1] A_Data[2] A_Data[3]	DTCMaskRecord[] = [DTCHighByte DTCMiddleByte DTCLowByte]	 C C C	 0x00~0xFF 0x00~0xFF 0x00~0xFF	DTCMREC_ DTCHB DTCMB DTCLB
A_Data[4]	DTCSnapshotRecordNumber	C	0x00~0xFF	DTCSSRN

Table 3 RDTCI Service request

(sub-function = reportDTCExtendedDataRecordByDTCNumber)

A_PDU Parameter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI	ReadDTCInformation Req ID	M	0x19	RDTCI
A_Data[0]	sub-function = [reportDTCExtendedDataRecord ByDTCNumber	M	0x06	LEV_ RDTCEDRB DN
A_Data[1] A_Data[2] A_Data[3]	DTCMaskRecord[] = [DTCHighByte DTCMiddleByte DTCLowByte]	 C C C	 0x00~0xFF 0x00~0xFF 0x00~0xFF	DTCMREC_ DTCHB DTCMB DTCLB
A_Data[4]	DTCExtendedDataRecordNumber	C	0x00~0xFF	DTCEDRN

Table 4 RDTCI Servie request (sub-function = reportSupportedDTC)

A_PDU Parameter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI	ReadDTCInformation Req ID	M	0x19	RDTCI
A_Data[0]	sub-function = [reportSupportedDTC]	M	0x0A	LEV_ RSUPDTC

3.3.10.3. Response messages

Table 1 RDTCI Positive response (reportNumberOfDTCByStatusMask)

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A_PDU Parameter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI	ReadDTCInformation Resp ID	M	0x59	RDTCIPR
A_Data[0]	reportType = [reportNumberOfDTCByStatusMask]	M	0x01	LEV_ RNODTCBSM
A_Data[1]	DTCStatusAvailabilityMask	M	0x00~0xFF	DTCSAM
A_Data[2]	DTCFormatIdentifier = [ISO15031-6DTCFormat ISO14229-1DTCFormat SAEJ1939-73DTCFormat]	M	0x00 0x01 0x02	DTCFID_ 15031-6DTCF 14229-1DTCF J1939-73DTCF
A_Data[3] A_Data[4]	DTCCount[] = [DTCCountHighByte DTCCountLowByte]	M M	0x00~0xFF 0x00~0xFF	DTCC_ DTCCHB DTCCLB

Table 2 RDTCI Positive response (reportDTCByStatusMask)

A_PDU Parameter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI	ReadDTCInformation Resp ID	M	0x59	RDTCIPR
A_Data[0]	reportType = [reportDTCByStatusMask]	M	0x02	LEV_ RDTCBSM
A_Data[1]	DTCStatusAvailabilityMask	M	0x00~0xFF	DTCSAM
A_Data[2] …… A_Data[m]	DTCAndStatusRecord[] = [DTCHighByte#1 DTCMiddleByte#1 DTCLowByte#1 statusOfDTC#1 DTCHighByte#2 DTCMiddleByte #2 DTCLowByte#2 statusOfDTC#2 …… DTCHighByte#m DTCMiddleByte#m DTCLowByte#m statusOfDTC#m ……]	C[1] C C C C[2] C C C …… C C C C ……	0x00~0xFF 0x00~0xFF 0x00~0xFF 0x00~0xFF 0x00~0xFF 0x00~0xFF 0x00~0xFF 0x00~0xFF …… 0x00~0xFF 0x00~0xFF 0x00~0xFF 0x00~0xFF ……	DTCASR_ DTCHB DTCMB DTCLB SODTC DTCHB DTCLB DTCFT SODTC …… DTCHB DTCMB DTCLB SODTC ……

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A_PDU Parameter	Parameter name	Cvt	Data(Hex)	Abbreviation
Note	[1] if reportType = reportDTCByStatusMask, and DTC information report required; [2] only if reportType = reportDTCByStatusMask, and more than 1 required DTC information report			

Table 3 RDTCI Positive response (reportDTCSnapshotIdentification)

A_PDU Parameter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI	ReadDTCInformation Resp ID	M	0x59	RDTCIPR
A_Data[0]	reportType = [reportDTCSnapshotIdentification]	M	0x03	LEV_ RDTCSSI
A_Data[1]	DTCRecord[] #1 = [DTCHighByte#1 DTCMiddleByte#1 DTCLowByte#1]	C[1] C C	0x00~0xFF 0x00~0xFF 0x00~0xFF	DTCASR_ DTCHB DTCMB DTCLB
A_Data[2]	DTCSnapshotRecordNumber #1	C	0x00~0xFF	DTCSSRN
……	……	……	……	……
……	DTCRecord[] #m = [DTCHighByte#m DTCMiddleByte#m DTCLowByte#m]	C[2] C C	0x00~0xFF 0x00~0xFF 0x00~0xFF	DTCASR_ DTCHB DTCMB DTCLB
……	DTCSnapshotRecordNumber #m	C	0x00~0xFF	DTCSSRN
Note	[1] at least 1 DTCSnapshot can be reported; [2] more than 1 DTCSnapshot record to be reported			

Table 4 RDTCI Positive response (reportDTCSnapshotRecordByDTCNumber)

A_PDU Parameter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI	ReadDTCInformation Resp ID	M	0x59	RDTCIPR
A_Data[0]	reportType = [reportDTCSnapshotRecordByDTCNumber]	M	0x04	LEV_ RDTCSSBDTC
A_Data[1] A_Data[2] A_Data[3] A_Data[4]	DTCAndStatusRecord[] = [DTCHighByte DTCMiddleByte DTCLowByte statusOfDTC]	M M M M	0x00~0xFF 0x00~0xFF 0x00~0xFF 0x00~0xFF	DTCASR_ DTCHB DTCMB DTCLB SODTC

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A_PDU Parameter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_Data[5]	DTCSnapshotRecordNumber #1	C[1]	0x00~0xFF	DTCSSRN
A_Data[6]	DTCSnapshotRecordNumberOfIdentifiers #1	C	0x00~0xFF	DTCSSRNI
A_Data[7] …… A_Data[m]	DTCSnapshotRecord[] #1 = [dataIdentifier#1 byte #1 (MSB) dataIdentifier#1 byte #2 snapshotData#1 byte #1 …… snapshotData#1 byte #p …… dataIdentifier#w byte #1 (MSB) dataIdentifier#w byte #2 snapshotData#w byte #1 …… snapshotData#w byte #m]	 C C C …… C …… C[2] C C …… C	 0x00~0xFF 0x00~0xFF 0x00~0xFF …… 0x00~0xFF …… 0x00~0xFF 0x00~0xFF 0x00~0xFF …… 0x00~0xFF	DTCSSR_ DIDB11 DIDB12 SSD11 …… SSD1p …… DIDB21 DIDB22 SSD21 …… SSD2m
……	……	……	……	…….
A_Data[t]	DTCSnapshotRecordNumber #x	C[3]	00~FF	DTCSSRN
A_Data[t+1]	DTCSnapshotRecordNumberOfIdentifiers #x	C	00~FF	DTCSSRNI
……	……	……	……	……
Note:	[1] at least 1 DTCSnapshot record can be reported; [2] 1 DTCSnapshot record includes multiple dataIdentifier/snapshotData; [3] all DTCSnapshot records need to be reported			

Table 5 RDTCI Positive response (reportDTCExtendedDataRecordByDTCNumber)

A_PDU Parameter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI	ReadDTCInformation Resp ID	M	0x59	RDTCIPR
A_Data[0]	reportType = [reportDTCSnapshotRecordByDTCNumber]	M	0x06	LEV_RDTCEDRBDN

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A_PDU Parameter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_Data[1] A_Data[2] A_Data[3] A_Data[4]	DTCAndStatusRecord[] = [DTCHighByte DTCMiddleByte DTCLowByte statusOfDTC]	 M M M M	 0x00~0xFF 0x00~0xFF 0x00~0xFF 0x00~0xFF	DTCASR_ DTCHB DTCMB DTCLB SODTC
A_Data[5]	DTCExtendedDataRecordNumber #1	C[1]	0x00~0xFF	DTCEDRN
A_Data[6] …… A_Data[m]	DTCExtendedDataRecord[] #1 = [extendedData#1 byte #1 (MSB) …… extendedData #1 byte #p	 C …… C	 0x00~0xFF …… 0x00~0xFF	DTCEDR_ EDD11 …… EDD1p
……	……	……	……	…….
A_Data[t]	DTCExtendedDataRecordNumber #x	C[2]	0x00~0xFF	DTCEDRN
A_Data[t+1] …… A_Data[t+1+(p+1)]	DTCExtendedDataRecord[] #x = [extendedData#x byte #1 (MSB) …… extendedData #x byte #p	 C …… C	 0x00~0xFF …… 0x00~0xFF	DTCEDR_ EDDx1 …… EDDxp
Note:	[1] DTCExtendedDataRecordNumber is not 0xFF，and the regarding extended data records can be accessible； Or DTCExtendedDataRecordNumber is 0xFF，at least 1 extended data record can be accessible； [2] DTCExtendedDataRecordNumber is 0xFF，more than 1 current record can be accessible；			

Table 6 RDTCI Positive response (reportSupportedDTCs)

A_PDU Parameter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI	ReadDTCInformation Resp ID	M	0x59	RDTCIPR
A_Data[0]	reportType = [reportSupportedDTCs]	M	0x0A	LEV_ RSUPDTC
A_Data[1]	DTCStatusAvailabilityMask	M	0x00~0xFF	DTCSAM

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A_PDU Parameter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_Data[2]	DTCAndStatusRecord[] = [			DTCASR_
……	DTCHighByte#1	C[1]	0x00~0xFF	DTCHB
	DTCMiddleByte#1	C	0x00~0xFF	DTCMB
	DTCLowByte#1	C	0x00~0xFF	DTCLB
	statusOfDTC#1	C	0x00~0xFF	SODTC
	DTCHighByte#2	C[2]	0x00~0xFF	DTCHB
	DTCMiddleByte #2	C	0x00~0xFF	DTCLB
	DTCLowByte#2	C	0x00~0xFF	DTCFT
	statusOfDTC#2	C	0x00~0xFF	SODTC
	……	……	……	……
	DTCHighByte#m	C	0x00~0xFF	DTCHB
	DTCMiddleByte#m	C	0x00~0xFF	DTCMB
	DTCLowByte#m	C	0x00~0xFF	DTCLB
	statusOfDTC#m]	C	0x00~0xFF	SODTC
注：	[1] only if reportType = reportSupportedDTCs，and there is DTC information report required； [2] only if reportType = reportSupportedDTCs，and more than 1 DTC information report required			

Table 7 RDTCI Negative response

A_PDU Parameter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI[0]	Negative Response ID	M	0x7F	SIDNR
A_PCI[1]	ReadDTCInformation Req ID	M	0x19	RDTCI
A_Data[0]	Negative Response Code	M	0x12, 0x13	NRC_

Table 8 RDTCI Negative response codes (NRC_)

Data(Hex)	Description	Cvt	Abbreviation
0x12	subFunctionNotSupported	M	SFNS
0x13	incorrectMessageLengthOrInvalidFormat	M	IMLOIF

Note:

For this diagnostic request , 2 sub-functions($04 and $06) are related to snap shot data record number as in the below :

$19 $04 $XX $YY $ZZ $PP or $19 $06 $XX $YY $ZZ $PP

(where $XXYYZZ represents the DTC code and $PP is Snap shot record number)

In current implementation, only one snap shot record for each DTC is supported and the snap shot record number support is 0x01.

So the last byte ($PP) in the diagnostic request to read the snapshot or extended snapshot data contains

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the snapshot record number. And if any diag request is sent with last byte value other than $01 or $FF, ECU will return a negative response.

3.3.11. InputOutputControlByIdentifier (IOCBI)(SID = 0x2F)

This service shall be available only if the following conditions are satisfied:

1) Vehicle speed < 3 Kmph
2) No Control Algorithm is Active
3) NO I/O Control failure condition is present!
4) At least one Wheel Speed Sensor is valid

3.3.11.1. Service Request

Table 1 IOCBI Service request

A_PDU Parameter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI[0]	InputOutputControlByIdentifier Request ID	M	0x2F	IOCBI
A_Data[0] A_Data[1]	dataIdentifier#1[] = [byte#1(MSB) byte#2(LSB)]	M M	0x00~0xFF 0x00~0xFF	IOI_ B1 B2
A_Data[2] …… A_Data[m+1]	controlOptionRecord#1[] = [controlState#1[inputOutputControlParameter[1]] …… controlState#m]	M …… C	0x00~0xFF …… 0x00~0xFF	CSR_ IOCP_/CS_ …… CS_
A_Data[m+2] …… A_Data[m+2+(r-1)]	controlEnableMaskRecord#1[] =[controlMask#1 …… controlMask#r]	C …… C	0x00~0xFF …… 0x00~0xFF	CEM_ CM_ …… CM_
Note:	[1] inputOutputControlParameter --- See Diagnostic Annex file			

3.3.11.2. Response messages

Table 1 IOCBI Positive response

A_PDU Parameter	Parameter name	Cvt	Data(Hex)	Abbreviation

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A_PDU Parameter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI[0]	InputOutputControlByIdentifier Response ID	S	0x6F	IOCBIPR
A_Data[2] A_Data[3]	dataIdentifier#1[] = [byte#1(MSB) byte#2(LSB)]	 M M	 0x00~0xFF 0x00~0xFF	IOI_ B1 B2
……	controlStatusRecord#1[] = [controlState#1/InputOutputControlParameter …… controlState#m：response data]	 C …… C	 0x00~0xFF …… 0x00~0xFF	CSR_ IOCP_/CS_ …… CS_

Table 2 IOCBI Negative response

A_PDU Parameter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI[0]	Negative Response ID	M	0x7F	SIDNR
A_PCI[1]	InputOutputControlByIdentifier Request ID	M	0x2F	IOCBI
A_Data[0]	Negative Response Code	M	0x13, 0x22, 0x31, 0x33	NRC_

Table 3 IOCBI Negative response codes (NRC_)

Data(Hex)	Description	Cvt	Abbreviation
0x13	incorrectMessageLengthOrInvalidFormat	M	IMLOIF
0x22	conditionNotCorrect	M	CNC
0x31	requestOutOfRange	M	ROOR
0x33	securityAccessDenied	M	SAD

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3.3.12. RequestDownload (RD) (SID = 0x34)

3.3.12.1. Service Request

Table 1 RD Service request

A_PDU Parameter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI[0]	RequestDownload Req ID	M	0x34	RD
A_Data[0]	dataFormatIdentifier	M	0x00~0xFF	DFI_
A_Data[1]	addressAndLengthFormatIdentifier	M	0x44	ALFID
A_Data[2] …… A_Data[5]	memoryAddress[] = [byte#1(MSB) …… byte#4]	M …… M	0x00~0xFF …… 0x00~0xFF	MA_B1 …… B4
A_Data[6] …… A_Data[9]	memorySize[] = [byte#1(MSB) …… byte#4]	M …… M	0x00~0xFF …… 0x00~0xFF	MS_B1 …… B4

Table 2 RD Service request parameter data definition

Name
dataFormatIdentifier： 1 byte: Higher 4 bits for“compressionMethod; Lower 4 bits for “encryptingMethod”. 0x00 means neither compressionMethod nore encryptingMethod is used. The other values are defined by supplier except for 0x00.
addressAndLengthFormatIdentifier： 7-4 bits: the length of memorySize 3-0bits: the length of memoryAddress
memoryAddress： Start address of the data memory. The data length is defined by 3-0 bits of addressFormatIdentifier.
memorySize (unCompressedMemorySize)： Server should use this parameter to compare the size of the transferred data and the size of the data memory.

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3.3.12.2. Response messages

Table 1 RD Positive response

A_PDU Parafmeter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI[0]	RequestDownload Resp ID	S	0x74	RDPR
A_Data[0]	lengthFormatIdentifier	M	0x00~0xF0	LFID
A_Data[1] …… A_Data[m]	maxNumberOfBlockLength = [byte#1 (MSB) …… byte#m]:	M …… M	0x00~0xFF …… 0x00~0xFF	MNROB B1 …… Bm

Table 2 RD Positive response data parameter definition

Parameter Name
lengthFormatIdentifier: 7-4 bits: the length of maxNumberOfBlockLength 3-0 bits: reserved, set all as 0
maxNumberOfBlockLength: The maximal data block legth of each TransferData request (maxNumberOfBlockLength)。The client side can use this parameter to be compatible to the receiving buffers of the server before transferring the data to the server.

Table 3 RD Negative response

A_PDU Parameter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI[0]	Negative Response ID	M	0x7F	SIDNR
A_PCI[1]	RequestDownload Req ID	M	0x34	RD
A_Data[0]	Negative Response Code	M	0x13, 0x22, 0x31, 0x33, 0x70	NRC_

Table 4 RD Negative response codes (NRC)

Data(Hex)	Description	Cvt	Abbreviation
0x13	incorrectMessageLengthOrInvalidFormat	M	IMLOIF
0x22	conditionNotCorrect	M	CNC

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Data(Hex)	Description	Cvt	Abbreviation
0x31	requestOutOfRange： 1、dataFormatIdentifier not correct； 2、memoryAddress/memorySize not correct 3、addressAndFormatIdentifier not correct ……	M	ROOR
0x33	securityAccessDenied	M	SAD
0x70	uploadDownloadNotAccepted	M	UDNA

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3.3.13. TransferData(TD) (SID = 0x36)

3.3.13.1. Service Request

Table 1 TD Service request

A_PDU Parameter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI[0]	TransferData Request ID	M	0x36	TD
A_Data[0]	blockSequenceCounter	M	0x00~0xFF	BSC
A_Data[1] …… A_Data[m]	transferRequestParameterRecord[] = [transferRequestParameter#1 …… transferRequestParameter#m]	U …… U	0x00~0xFF …… 0x00~0xFF	TRPR_ TRTP_ …… TRTP_

3.3.13.2. Response messages

Table 2 TD Positive response

A_PDU Parameter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI[0]	TransferData Response ID	S	0x76	RUPR
A_Data[0]	blockSequenceCounter： Not compatible to the value in request messages	M	00~FF	BSC
A_Data[1] …… A_Data[n]	transferResponseParameterRecord[] = [transferResponseParameter#1 …… transferResponseParameter#m]：	M …… M	00~FF …… 00~FF	TREPR_ TREP_ …… TREP_

Table 3 TD Request data and positive response data parameters definition

Parameter name
blockSequenceCounter： The blockSequenceCounter of the first TransferData request is 0x01, then this value of the next TransferData request should increase by 1. When this value reaches to 0xFF， the next value of the next TransferData should be started from 0x00. The parameter is used for the extended error handling.

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Parameter name
transferRequestParameterRecord: The transferRequestParameterRecord includes the parameters which the server needs for supporting the datat transferring. The format and the lemgth of the paramters are decided by the supplier. For the download data, the transferRequestParameterRecord includes the transferred data.
transferResponseParameterRecord: The transResponseParameterRecord includes the paramteters which the client needs for supporting the transferred data. The format and the length of the paramters are decided by the supplier. While downloading the data, these parameters include the Checksum calculated by the server. Note : TransferResponseParameterRecord is not repeating the transferRequestParameterRecord.

Table 4 TD Negative response

A_PDU Parameter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI[0]	Negative Response ID	M	0x7F	SIDNR
A_PCI[1]	TransferData Request ID	M	0x36	TD
A_Data[0]	Negative Response Code	M	0x13, 0x24, 0x31, 0x71, 0x72, 0x73, 0x92/0x93	NRC_

Table 5 TD Negative response codes (NRC_)

Data(Hex)	Description	Cvt	Abbreviation
0x 13	incorrectMessageLengthOrInvalidFormat	M	IMLOIF
0x 24	requestSequenceError: 1、While ECU receiving the service request, RequestDownload or RequestUpload service is not activated; 2、While ECU receiving the service request, RequestDownload or RequestUpload service is activated, but all the transferred data have been transmitted completely.	M	RSE

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Data(Hex)	Description	Cvt	Abbreviation
0x 31	requestOutOfRange If transferRequestParameterRecord includes the additional parameter but it is not correct.	M	ROOR
0x 71	transferDataSuspended	M	TDS
0x 72	generalProgrammingFailure	M	GPF
0x 73	wrongBlockSequenceCounter When server detects the sequence error of the blockSequenceCounte. Note： When ECU receives the data transferring requests with the same blockSequenceCounter continuously, it should receive the requests normally. ECU should give the positive response but not write the same data into memory repeatedly during the data downloading; but ECU should repeat upload the data during the data uploading.	M	WBSC
0x 92/93	Voltage too high/Voltage too low	M	VTH/VTL

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3.3.14. RequestTransferExit (RTE) (SID = 0x37)

3.3.14.1. Service Request

Table 1 RTE Service request

A_PDU Parameter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI[0]	Request TransferExit Req ID	M	0x37	RTE
A_Data[1] …… A_Data[m]	transferRequestParameterRecord[] = [transferRequestParameter#1 …… transferRequestParameter#m]	U …… U	0x00~0xFF …… 0x00~0xFF	TRPR_ TRTP_ …… TRTP_

3.3.14.2. Response messages

Table 2 RTE Positive response

A_PDU Parameter	Parameter name	Cvt	Data (Hex)	Abbreviation
A_PCI[0]	Request TransferExit Response ID	M	0x77	RTEPR
A_Data[1] …… A_Data[n]	transferResponseParameterRecord[] = [transferResponseParameter#1 …… transferResponseParameter#m]	U …… U	00~FF …… 00~FF	TREPR_ TREP_ …… TREP_

Table 3 RTE Negative response

A_PDU Parameter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI[0]	Negative Response ID	M	0x7F	SIDNR
A_PCI[1]	RequestTransferExit Request ID	M	0x37	RTE
A_Data[0]	Negative Response Code	M	0x13, 0x22, 0x24	NRC_

Table 4 RTE Negative response codes (NRC_)

Data(Hex)	Description	Cvt	Abbreviation
0x 13	incorrectMessageLengthOrInvalidFormat	M	IMLOIF
0x 22	conditionsNotCorrect While receiving the service request, the programming has not been finished.	M	CNC

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Data(Hex)	Description	Cvt	Abbreviation
0x 24	requestSequenceError: RequestDownload or RequestUpload is not active, but the RequestDownload or RequestUpload is requested.	M	RSE

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3.3.15. RoutineControl (RC) (SID = 0x31)

The Start Routine service shall be available only if the following conditions are satisfied:

1) Vehicle speed < 3kmph
2) No Control Algorithm is Active
3) At least one Wheel Speed Sensor is valid

3.3.15.1. Service Request

Table 1 RC Service request

A_PDU Parameter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI[0]	RoutineControl Request ID	M	0x31	RC
A_Data[0]	sub-function[] = [routineControlType]	M	0x00~0xFF	LEV_ RCTP_
A_Data[1] A_Data[2]	routineIdentifier[] = [byte1(MSB) byte2]	M M	0x00~0xFF 0x00~0xFF	RI_ B1 B2
A_Data[3] …… A_Data[n]	routineControlOptionRecord[] = [routineControlOption#1 …… routineControlOption#m]	C/U[1] …… C/U	0x00~0xFF …… 0x00~0xFF	RCEOR_ RCO_ …… RCO _
Note	[1] C/U is valid only when sub-function is stopRoutine or startRoutine.			

3.3.15.2. Response messages

Table 2 RC Positive response

A_PDU Parameter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI[0]	RoutineControl Response ID	M	0x71	RCPR
A_Data[0]	routineControlType	M	0x00~0xFF	RCTP_
A_Data[1] A_Data2]	routineIdentifier [] = [byte#1 (MSB) byte#2]	M M	0x00~0xFF 0x00~0xFF	RI_ B1 B2
A_Data[3] …… A_Data[n]	routineStatusRecord[] = [routineStatus#1 …… routineStatus#m]	U …… U	0x00~0xFF …… 0x00~0xFF	RSR_ RS_ …… RS _

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Table 3 RoutineControlType(LEV_ RCTP_) data definition

Data(Hex)	Description	Cvt	Abbreviation
0x 01	startRoutine This parameter is used for the regarding routine start of the server.	M	STR
0x 02	stopRoutine This parameter is used for stopping the reqgarding rountine of the server.	M	STPR
0x 03	requestRoutineResults	M	RRR

Table 4 RC Negative response

A_PDU Parameter	Parameter name	Cvt	Data(Hex)	Abbreviation
A_PCI[0]	Negative Response ID	M	0x7F	SIDNR
A_PCI[1]	RoutineControl Request ID	M	0x31	RC
A_Data[0]	Negative Response Code	M	0x12, 0x13, 0x22, 0x24, 0x31, 0x33, 0x72	NRC_

Table 5 RC Negative response codes (NRC_)

Data(Hex)	Description	Cvt	Abbreviation
0x 12	subFunctionNotSupported	M	SFNS
0x 13	incorrectMessageLengthOrInvalidFormat	M	IMLOIF
0x 22	conditionsNotCorrect	M	CNC
0x 24	requestSequenceError Have to firstly 'startRoutine' , then 'stopRoutine' or 'requestRoutineResults'	M	RSE
0x 31	requestOutOfRange Servier doesn't support the requested routineIdentifier or routineControlOptionRecord includes the invalid data.	M	ROOR
0x 33	securityAccessDenied	M	SAD
0x 72	GeneralProgrammingFailure For example, the server finds the failure when the memory of the server is accessed.	M	GPF

Table 6 routineIdentifier definition

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Data(Hex)	Description	Cvt	Abbreviation
0x FF00	eraseMemory	M	EM
0x FF01	checkProgrammingDependencies	M	CPD

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4. Supported Identifiers

This section defines the identifiers (IDs) that are to be supported for the service ReadDataByIdentifier (SID 22), WriteDataByIdentifier (SID 2E) and IOControlByIdentifier (SID 2F).

Identifiers may be used to read and specify ECU status and I/O states as well as internal constants or variables.

The Services in which each identifier is supported is specified in the Columns title “Accessed Via”.

The Services are identified by their hexadecimal SID (Service Identifier), as per ISO 14229. The security access level require to access the identifier is specified by a numeric value.

Value	Security Access Level
0	No Security Access Required
1	Extended Security Level
X	Programming Security Level
X	Other Security Level

Table 4-1: Identifier Access Levels

Note: In CHANA CD101 ABS/VSC projects, only Security Access Level 1 and its logic algorithm for Extended Security Level is defined officially. The other Security Access Level (for example Programming Security Level or Supplier Security Level and its logic algorithm) can be defined by TRW if needed according to the requirement from CHANA.

For example, in the table below, the identifier (1234) can be read without performing security access (readDataByIdentifier, 22), but is protected by the extended security access for IOControl Service (IOControlByIdentifier, 2F). In this example, it is not possible to write data to the identifier (WriteDataByIdentifier, 2E)

Note. If a Diagnostic Service is not supported, there will be no security access level specified.

ID No	ID Name	Accessed via			Byte	Bit	Signal/ Variable name	Bit Description	0	1
		22	2E	30						
1234	Sample Id	✓0	☒	✓1	1	0	-	Sample bit 0	off	on
						1	-	Sample bit 1	off	on
						2	-	Sample bit 2	off	on
						3	-	Sample bit 3	off	on
						4	-	Sample bit 4	off	on
						5	-	Sample bit 5	off	on
						6	-	Sample bit 6	off	on
						7	-	Sample bit 7	off	on

Table 4-2: Sample Identifier table

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4.1. Bitmapped Identifiers

Define all bitmapped identifiers in this section.

ID No	ID Name	Accessed via 22	2E	2F	Byte	Bit	Signal/ Variable name	Bit Description	0	1
D300	Digital Status	✓0	☒	☒	1	7	WheelSpeedFLFault	Front Left wheel speed fault	No fault	fault
						6	WheelSpeedFRFault	Front Right wheel speed fault	No fault	fault
						5	WheelSpeedRLFault	Rear Left wheel speed fault	No fault	fault
						4	WheelSpeedRRFault	Rear Right wheel speed fault	No fault	fault
						3	SteeringAngleValid(VSC Only) 0 for ABS	Steering Angle Valid signal	Not valid	Valid
						2	YawRateValid(VSC Only) 0 for ABS	Yaw Rate valid signal	Not valid	Valid
						1	VehicleSpeedValid	vehicle speed valid	Not valid	Valid
						0	WheelRoughRoadValid	rough road signal valid	Not valid	Valid
					2	7	ESPTorqRequestAvailability(VSC Only) 0 for ABS	ASC acknowledge	No fault	fault
						6	EngineCoolantTempFault(VSC Only) 0 for ABS	engine temperature sensor fault	No fault	fault
						5	EngineSpeedFault (VSC only)	engine speed error	No fault	fault
						4	PedalPosnAccelFault(VSC Only) 0 for ABS	sensor fault with accelerator pedal position	No fault	fault
						3	TorqueIndicatedFault(VSC Only) 0 for ABS	torque measurement error	No fault	fault
						2	BrakeLightSwitchValid	brake light switch fault	No fault	fault
						1	reserved	reserved	-	-
						0	reserved	reserved	-	-
					3	7	reserved	reserved	-	-
						6	reserved	reserved	-	-
						5	BrakeLightSwitch	brake light switch	off	on
						4	KeySwitchStateIGN(VSC Only) 0 for ABS	key position 2/IGN	off	on
						3	ReverseGear(VSC Only) 0 for ABS	FGP signal - reverse gear	off	on
						2	Reserved	Reserved	off	on
						1	TC/VSCStatus(VSC Only) 0 for ABS	TC/VSC intervention	off	on
						0	CruiseControlActive(VSC Only) 0 for ABS	Cruise control active. This bit is not implemented in CD101/C201 and reserved for future platform development.	off	on
					4	7	GearShiftActive(VSC Only) 0 for ABS	gear shift in progress	off	on

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						6	TorqueIncMSRActive(VSC Only) 0 for ABS	torque increase MSR active This bit is not implemented in CD101/C201 and reserved for future platform development.	off	on
						5	TorqueRedASRActive(VSC Only) 0 for ABS	torque reduction ASR active	off	on
						4 3	Driver Mode Switch (VSC only); 0 for ABS	Driver Mode Switch Status	00 TC/VSC OFF 01 TC/VSC ON 02 Unused 03 Invalid	
						2	Reserved	reserved	-	-
						1	Reserved	reserved	-	-
						0	reserved	reserved	-	-

Note: I/O control will be inactive if the vehicle speed exceeds 3 kph.

ID No	ID Name	Accessed via 22	2E	2F	Byte	Bit	Signal/ Variable name	Bit Description	0	1
D301	IO Control Identifiers	✓0	☒	✓1	1	7	reserved	reserved		
						6	reserved	reserved		
						5		Pump motor	off	on
						4		FS relay. Short term adjustment via IO control is not allowed.	off	on
						3		reserved		
						2		TC/VSC info lamp (VSC Only) 0 for ABS	off	on
						1		DRP w.lamp	off	on
						0		ABS w.lamp	off	on
					2	7		RR iso	off	on
						6		RR dump	off	on
						5		LR iso	off	on
						4		LR dump	off	on
						3		RF iso	off	on
						2		RF dump	off	on
						1		LF iso	off	on
						0		LF dump	off	on
					3	7	reserved	reserved		
						6	reserved	reserved		
						5	reserved	reserved		
						4	reserved	reserved		
						3		RFLR TC iso(VSC Only) 0 for ABS	off	on
						2		LFRR TC iso (VSC Only) 0 for ABS	off	on
						1		RFLR TC supply (VSC Only) 0 for ABS	off	on

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						0		LFRR TC supply (VSC Only) 0 for ABS	off	on
					4	7	reserved	reserved		
						6	reserved	reserved		
						5	reserved	reserved		
						4	reserved	reserved		
						3	reserved	reserved		
						2	reserved	reserved		
						1	reserved	reserved		
						0	reserved	reserved		

Table 4-3: Bitmapped Identifier table

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4.2. Non Bitmapped Identifiers

ID No	Short Name (or Signal / variable Name)	Accessed via 22	2E	2F	Size (bits)	Description	Resolution	Offset	Min	Max.
0112	Battery Voltage Identifier	✓0	☒	☒	16	FFFF = Undefined	1 bit = 1/256V	0	0	FFFE
6300	Hydraulic Filling Indicator	✓ 0	✓ 1	☒	8	Filling not successfully = 00 Filling successfully = 7F Delivery State = FF				
6302(VSC Only)	Read Pressure	✓0	☒	☒	16		1bit = 1/16 bar		0	FFFE
6304(VSC Only)	Read Steering Angle Flag	✓0	☒	☒	8	Returns the calibration status of the SAS sensor.00 = Not Calibrated 01 = Calibrated				
6310	WheelSpeedFL (kph)	✓0	☒	☒	13	Front Left wheel speed CAN signal	0.05625	0	0	0x1900
6311	WheelSpeedFR (kph)	✓0	☒	☒	13	Front Right wheel speed CAN signal	0.05625	0	0	0x1900
6312	WheelSpeedRL (kph)	✓0	☒	☒	13	Rear Left wheel speed CAN signal	0.05625	0	0	0x1900
6313	WheelSpeedRR(kph)	✓0	☒	☒	13	Rear Right wheel speed CAN signal	0.05625	0	0	0x1900
6321(VSC Only)	TC/VSCStatus	✓0	☒	☒	2	TC/VSC status CAN signal 0 = off 1 = intervention 2 = fail 3 = ok				
6340(VSC Only)	SteeringAngle	✓0	☒	☒	16	Steering Angle signal ,Value type is signed, The definition is same with the CAN signal in message 0x180	1bit = 0.1deg	0	-780	780

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ID No	Short Name (or Signal / variable Name)	Accessed via 22	2E	2F	Size (bits)	Description	Resolution	Offset	Min	Max.
6344(VSC Only)	YawRate	✓0	☒	☒	16	Yaw Rate signal of IIS	1bit = 1/16deg/s	-128deg/s	0	FFFE
6346(VSC Only)	Read Lateral Acceleration	✓0	☒	☒	16	Accelerate Signal of IIS	1bit =1/64m/sec²	-32 m/sec²	0	FFFE
010F	VehicleSpeed	✓0	☒	☒	16	vehicle speed CAN Signal	0.05625	0	0	0x1900

Table 4-4: Non Bitmapped Identifier table

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4.3. Logistic Data

ID	NAME	Supported By				Memory Area	Single Write[1]		Erase on Reprogramming	Data		Additional Info
		Applicatio		Boot								
		22	2E	22	2E		Supplier	CHANA		Size(Bytes)	Type	
F184	Application Software Fingerprint Data Identifier (Confirmed by CHANA)	✓0	✓1	☒	☒	NVRAM	N	N	N	12	MUX	Byte 0…6: Tool Supplier Identification (ASCII) Byte 7: Programming Year (0x00: Year 2000; 0x01: Year 2001……) (Unsigned) Byte 8: Programming Month (0x01: January; …0x0C : December) (Unsigned) Byte 9: Programming Data(0x01: 1stst;st… 0x1F: 31^{s}) (Unsigned) Byte 10,11: Test Serial Number (Unsigned)
F185	Calibration Software (Application Data) Fingerprint Data Identifier. For TRW ABS/ESP: the same as F184 as the Application SW and the Calibration SW are integrated. (Confirmed by CHANA)	✓0	✓1	☒	☒	NVRAM	N	N	N	12	MUX	

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ID	NAME	Supported By				Memory Area	Single Write[1]		Erase on Reprogramming	Data		Additional Info
		Applicatio		Boot								
		22	2E	22	2E		Supplier	CHANA		Size(Bytes)	Type	
F120	CHANA Network Protocol Version Number (Confirmed by CHANA)	✓0	☒	☒	☒	ROM	N	N	N	4	ASCII	**CHANA Network Protocol Version is used for ECU CAN configuration traceability. For example, V1.0 CAN Protocol is used.** **(Confirmed by CHANA)**
F187	CHANA Part Number (Assembly Number) (Confirmed by CHANA)	✓0	✓1	☒	☒	NVRAM	N	N	Y	10	ASCII	The Assembly number will be updated by the tester following reprogramming. The assembly number is a unique part number that uniquely identifies the collection of software files and Hardware files. This identifier must be updated by the tester. Download is performed that results in a software module changing. To be written by Supplier / CHANA.

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ID	NAME	Supported By				Memory Area	Single Write[1]		Erase on Reprogramming	Data		Additional Info
		Applicatio		Boot								
		22	2E	22	2E		Supplier	CHANA		Size(Bytes)	Type	
F190	Vehicle Identification Number (Confirmed by CHANA)	✓0	✓1	☒	☒	NVRAM	N	Y	N	17	ASCII	The VIN number of the vehicle that the ECU is fitted to. The identifier is written to when a component is first fitted to a vehicle, either Manufacturing or in Service as a replacement part. Once written to, it shall **not** be possible to overwrite this data by reprogramming or service 2E. Written BY CHANA only. This is the 17 characters of the full VIN in ASCII (Confirmed by CHANA)
F193	CHANA ECU Hardware Version Number (Confirmed by CHANA)	✓0	☒	☒	☒	NVRAM	Y	N	N	10	ASCII	The Part Number of the ECU as assigned by the CHANA Bill of Material (BOM). The Part Number shall be written during the ECU manufacturing process and shall not be modified during the life of the ECU. **Written BY SUPPLIER only.**

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ID	NAME	Supported By				Memory Area	Single Write[1]		Erase on Reprogramming	Data		Additional Info
		Applicatio		Boot								
		22	2E	22	2E		Supplier	CHANA		Size(Bytes)	Type	
F195	CHANA ECU Software Version Number Data Identifier (Confirmed by CHANA)	✓0	☒	☒	☒	ROM	N	N	Y	10	ASCII	This item should be populated automatically by the ECU during the reprogramming process. The Software part number shall be embedded in the ECU software file. **This number will be release in CHANA Bill of Material. i.e 1234567801 is "12345678/ 01".**
F18A	Supplier ID Number (Confirmed by CHANA)	✓0	☒	☒	☒	ROM	Y	N	N	7	ASCII	Seven bytes Supplier Id Numbers will be released to TRW, which will be stored in application SW (ROM). CHANA provides the content of these bytes to TRW.

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ID	NAME	Supported By				Memory Area	Single Write[1]		Erase on Reprogramming	Data		Additional Info
		Applicatio		Boot								
		22	2E	22	2E		Supplier	CHANA		Size(Bytes)	Type	
F18B	Supplier ECU Manufacture Date (Date of Final Test) (Reserved for TRW plant) (To be discussed with CHANA)	✓0	☒	☒	☒	NVRAM	Y	N	N	3	BCD	The Data of Manufacture of the ECU. Written during the ECU manufacturing process **Written By SUPPLIER Only** **i.e 070815 is 15th August 2007.**
F18C	Supplier ECU Serial Number (Confirmed by CHANA)	✓0	☒	✓0	☒	NVRAM	Y	N	N	10	HEX	A serial number that identifies the ECU uniquely. Format and content may be defined by supplier. **Written By SUPPLIER Only** **TRW Internal format to be used as discussed with CHANA.**

Table 4-5: Logistic Identifier table

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4.4. Configuration Data

Identifier	Name	22	2E	Byte No	Bit No	Variable Name	Variable Description
C101 (Only used for VSC Car Configuration)	CHANA CD101 VSC Car Configuration	✓0	✓1	1		Gear Box Type (YSC only)	GEAR 6AT 00 GEAR_VAR_UNDEFINED 01
				2		Engine Type: (YSC only)	ENGINE 1.8T 00 ENGINE_VAR_UNDEFINED 01
				3		Brakes (YSC only) (reserved)	BRAKES_UNDEFINED 00
				4		Chassis (reserved)	CHASSIS_UNDEFINED 00
				5		Steering (YSC only) (reserved)	STEERING_UNDEFINED 00
				6		Wheel circ high byte (reserved)	High byte of size in 1 bit / mm (00)
				7		Wheel circ low byte (reserved)	Low byte of size in 1 bit / mm (00)
				8		***reserved***	Not Used 00
				9		***reserved***	Not Used 00
				10		***reserved***	Not Used 00
				11		***reserved***	Not Used 00
				12		***reserved***	Not Used 00

Table 4-6: Configuration data

Note:

Currently,CD101 contains only two engine type :1.8T and 2.0.
For 2.0 engine type ,ABS only is fitted.
For 1.8T engine type,ESC is fitted.
For all the variant of vehicle, EPB are all supported.

	CD101 2.0 MT	CD101 2.0 AT	CD101 1.8T AT
ABS only	X	X	
VSC			X
EPB supported	X	X	X

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4.4.1. Car configuration procedure:

1) Enter the corresponding diagnostic session

   The car configuration diagnostic service is allowed in Extended diagnostic session or in EOL diagnostic session (which is 10 03 and 10 04).
2) Enter the corresponding SECURITY ACCESS

   Please refer to CHANA's SECURITY ACCESS logic proposed by TRW.
3) The car configuration data are written into VSC ECU by Writing Data diagnostic service.

   For example, the car configuration data for CD101 ESC Car (Engine type 1.8T, Gear box type 6AT) should be: 2E C1 01 00 00 00 00 00 00 00 00 00 00 00 00

4.4.2. Variant Coding Fault detection

TRW ESC will check the written engine type configuration data comparing with the receiving engine configuration signal value on the CAN bus (the Engine type on the CAN bus from EMS 0x355) .If mismatch, 'variant coding mismatch' fault will be logged.

4.5. EOL diagnostic session for EOL at High speeds

4.5.1. EOL diagnostic session for EOL at High speeds

EOL session: EOL session (10 04) disables the ECU's control algorithms and provides additional functionality such as I/O control at high speed. The diagnostic subsystem has full control over the system, session timers are required.

The ECU shall switch to the requested EOL Session only if the following conditions are satisfied:

3) Vehicle speed < 3 kmph
4) No Control Algorithm is Active

After entering into the EOL session we have additional feature of performing IO control at high speeds by disabling the speed limit using write data by identifier (7216).

The procedure for IO control at High speeds:

EOL session→ Security Access → Disable speed limit using write data by identifier → IO control at high speeds.

Note: The Disable speed limit will be again enabled when it exits EOL session.

4.5.2. Disable/Enable Speed Limit Check

1）Write the Speed Limit State: SID$2E $7216

Data Record: 1 Byte

Allowed DIAG Sessions: EOL

Security Access: Required Security Level 1

Service does not support functional addressing.

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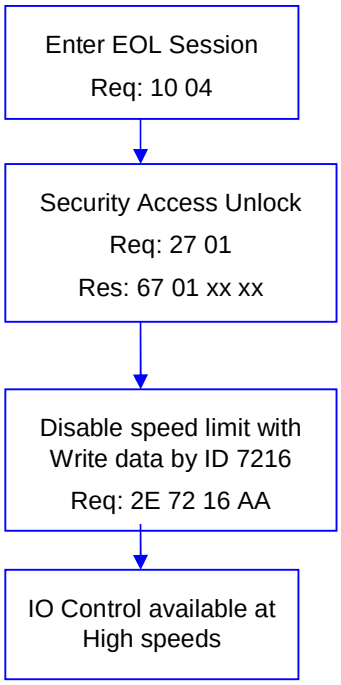
SID $2E, Request Message format				
Byte	**CAN ID**	**Name**		**Value**
#1	0x780	SID-Request	Mandatory	0x2E
#2	0x780	Speed Limit Disable DID (high byte)	Mandatory	0x72
#3	0x780	Speed Limit Disable DID (Low byte)	Mandatory	0x16
#4 to #20	0x780	Speed Limit Disable Value	Mandatory	0xAA – Disable Speed limit 0x00 – Enable speed limit

Positive Response (PR) Message format:				
Byte	**CAN ID**	**Name**		**Value**
#1	0x788	SID-Positive Response	Mandatory	0x6E
#2	0x788	Speed Limit Disable DID (high byte)	Mandatory	0x72
#3	0x788	Speed Limit Disable DID (Low byte)	Mandatory	0x16

2）The procedure for IO control at High speeds:

EOL session→ Security Access → Disable speed limit using write data by identifier → IO control at high speeds.



Note: The Disable speed limit will be again enabled when it exits EOL session.

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4.5.3. EOL Test by Routine Control

Routine ID	ID Name	Routine Control Type	Byte	Bit	Signal/ Variable name	Bit Description	0	1
3028	EOL test by Routine control (Note: The next state activation will remain 4 seconds if it is not stopped.)	01: Start Routine 02: Stop Routine 03: Request Results	1	7	Valves Actuation Current State	RR iso	off	on
				6		RR dump	off	on
				5		LR iso	off	on
				4		LR dump	off	on
				3		RF iso	off	on
				2		RF dump	off	on
				1		LF iso	off	on
				0		LF dump	off	on
			2	7	Valves and Pump Motor Actuation Current State	reserved		
				6		reserved		
				5		reserved		
				4		Pump motor	off	on
				3		RFLR TC iso(VSC Only) 0 for ABS	off	on
				2		LFRR TC iso (VSC Only) 0 for ABS	off	on
				1		RFLR TC supply (VSC Only) 0 for ABS	off	on
				0		LFRR TC supply (VSC Only) 0 for ABS	off	on
			3		Waiting time (High byte)	Range: 0x000 – 0x01FF 0ms – 5110ms Steps: 1 bit = 10 ms; Resolution: 10 ms		
			4		Waiting time (Low byte)			
			5	7	Valves Actuation Next State	RR iso	off	on
				6		RR dump	off	on
				5		LR iso	off	on
				4		LR dump	off	on
				3		RF iso	off	on
				2		RF dump	off	on
				1		LF iso	off	on
				0		LF dump	off	on
			6	7	Valves and Pump Motor Actuation Next State	reserved		
				6		reserved		
				5		reserved		
				4		Pump motor	off	on
				3		RFLR TC iso(VSC Only) 0 for ABS	off	on
				2		LFRR TC iso (VSC Only) 0 for ABS	off	on

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				1		RFLR TC supply (VSC Only) 0 for ABS	off	on
				0		LFRR TC supply (VSC Only) 0 for ABS	off	on

Table 4-7: Rotine ID

For example:

To create a pressure increasing at FL for 70ms the command:

31 01 30 28 A8 15 00 07 AA 15 is activating (01 is Start the routine; 02 is Stop the routine):

ABS iso valves FR,RL,RR --- A8

TC iso FLRR, TC supply FLRR and Pump motor --- 15

Waiting time --- 00 07

All ABS iso valves --- AA

TC iso FLRR, TC supply FLRR and Pump motor --- 15

4.6. Support Routines

The following Routines are required to be supported for use with the RoutineControl Service.

4.6.1. Application Software

ROUTINE ID ROUNTINEIDENTIFIER	FUNCTION	DESCRIPTION	ROUTINECONTROLOPTIONS [ROUTINECONTROLOPTION#1 : ROUTINECONTROLOPTION#M]	ROUTINECONTROLTYPE STARTROUTINE 01	STOPROUTINE 02	REQUESTRESULTS 03
3020	Master Cylinder Pressure Offset Determination (VSC Only)	Pressure sensor offset determination		✔ 1	x	✔ 1
3021	IIS Sensors Offset Determination (VSC Only)	EBC460 IIS sensors offset determination		✔ 1	x	✔ 1

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3027	Steering wheel angle offset Determination (VSC Only)	Steering angle sensor offset determination		✔ 1	X	✔ 1
0202	Demand on Self-test	Demand on system self-test	NA	✔ 1	X	✔ 1
3028	EOL Test by routine control (ABS and VSC)	EOL test by routine control for EBC460 ABS and VSC		✔ 1	✔ 1	✔ 1

Table 4-8: Supported Routines in application SW

The routine control results response of the request $31 $03 $XX $YY for all the routine control Ids in CHANA:

For System self test and EOL routine, the response data byte will be

0x00 - Routine completed

0x01 - Routine Running

For Sensor Calibration results, the response will be $71 $03 $30 $21 $XX

$XX: $00 – Routine successful

$01 – Routine running

$02 $YY $ZZ –Routine failed, where $YY gives the sensor ID and $ZZ gives the failure type

$YY: Sensor decoding

$01 – Yawrate sensor

$02 – Lateral sensor

$ZZ : failure type decoding

$01 – Signal not valid

$02 – Sensor signal out of range

$03 – Internal fault

In case sensor calibration fails,

i. In case of IIS calibration two additional bytes $YY $ZZ will be sent in the response.

ii. In case of MCP calibration only one addtional byte $ZZ is sent.

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4.6.2. Boot Software

<table>
<tr><th rowspan="2">ROUTINE ID</th><th rowspan="2">FUNCTION</th><th rowspan="2">DESCRIPTION</th><th rowspan="2">ROUTINECONTROLOPTIONS
[ROUTINECONTROLOPTION#1
:
ROUTINECONTROLOPTION#M]</th><th colspan="3">ROUTINECONTROLTYPE</th></tr>
<tr><th>STARTROUTINE 01</th><th>STOPROUTINE 02</th><th>REQUESTRESULTS 03</th></tr>
<tr><td>FF00</td><td>eraseMemory</td><td>Erase memory contents</td><td><table>
<tr><th>Request Parameter</th><th>Description
FF00</th></tr>
<tr><td>routineControlOption#1</td><td>addressAndLengthFormatIdentifier</td></tr>
<tr><td>routineControlOption#2
:
routineControlOption#n</td><td>memoryAddress</td></tr>
<tr><td>routineControlOption#n+1
:
routineControlOption#k</td><td>memorySize</td></tr>
</table></td><td>✓ 2</td><td>x</td><td>✓ 2</td></tr>
</table>

Table 4-9: Supported Routines in Boot SW

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5. TRW Supported DTCs List for CD101 Project

Note: Some DTCs are changed as per CHANA DTC request (ABS/ESP DTCs range: C1000 – C13FF; ABS/ESP Network DTCs range: U2100 – U217F defined by CHANA).

DESCRIPTION FOR SYMBOLS
☺ Fault Present

5.1. Supported DTC lists

No.	Description	Fault Byte-1	Fault Byte-2	Fault Byte-3	ABS	ESC	EBD Lamp	ABS Lamp	ESC Lamp
1	Wheel Speed Sensor Ohmic Open Failure - Front Left	0x50	0x35	0x13	☺	☺	OFF	ON	ON
2	Wheel Speed Sensor Ohmic Open Failure - Front Right	0x50	0x40	0x13	☺	☺	OFF	ON	ON
3	Wheel Speed Sensor Ohmic Open Failure - Rear Left	0x50	0x45	0x13	☺	☺	OFF	ON	ON
4	Wheel Speed Sensor Ohmic Open Failure - Rear Right	0x50	0x50	0x13	☺	☺	OFF	ON	ON
5	Wheel Speed Sensor Ohmic Short Failure - Front Left	0x50	0x35	0x12	☺	☺	OFF	ON	ON
6	Wheel Speed Sensor Ohmic Short Failure - Front Right	0x50	0x40	0x12	☺	☺	OFF	ON	ON
7	Wheel Speed Sensor Ohmic Short Failure - Rear Left	0x50	0x45	0x12	☺	☺	OFF	ON	ON
8	Wheel Speed Sensor Ohmic Short Failure - Rear Right	0x50	0x50	0x12	☺	☺	OFF	ON	ON
9	Wheel Speed Sensor Missing - Front Left	0x50	0x36	0x02	☺	☺	OFF	ON	ON
10	Wheel Speed Sensor Missing- Front Right	0x50	0x41	0x02	☺	☺	OFF	ON	ON
11	Wheel Speed Sensor Missing - Rear Left	0x50	0x46	0x02	☺	☺	OFF	ON	ON
12	Wheel Speed Sensor Missing - Rear Right	0x50	0x51	0x02	☺	☺	OFF	ON	ON
13	Wheel Speed Sensor Fast Missing - Front Left	0x50	0x36	0x02	☺	☺	OFF	ON	ON
14	Wheel Speed Sensor Fast Missing - Front Right	0x50	0x41	0x02	☺	☺	OFF	ON	ON

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No.	Description	Fault Byte-1	Fault Byte-2	Fault Byte-3	ABS	ESC	EBD Lamp	ABS Lamp	ESC Lamp
15	Wheel Speed Sensor Fast Missing - Rear Left	0x50	0x46	0x02	☺	☺	OFF	ON	ON
16	Wheel Speed Sensor Fast Missing - Rear Right	0x50	0x51	0x02	☺	☺	OFF	ON	ON
17	Wheel Speed Sensor Erratic - Front Left	0x50	0x36	0x2F	☺	☺	OFF	ON	ON
18	Wheel Speed Sensor Erratic - Front Right	0x50	0x41	0x2F	☺	☺	OFF	ON	ON
19	Wheel Speed Sensor Erratic - Rear Left	0x50	0x46	0x2F	☺	☺	OFF	ON	ON
20	Wheel Speed Sensor Erratic - Rear Right	0x50	0x51	0x2F	☺	☺	OFF	ON	ON
21	Wheel Speed Sensor Dropout Failure - Front Left	0x50	0x36	0x04	☺	☺	OFF	ON	ON
22	Wheel Speed Sensor Dropout Failure - Front Right	0x50	0x41	0x04	☺	☺	OFF	ON	ON
23	Wheel Speed Sensor Dropout Failure - Rear Left	0x50	0x46	0x04	☺	☺	OFF	ON	ON
24	Wheel Speed Sensor Dropout Failure - Rear Right	0x50	0x51	0x04	☺	☺	OFF	ON	ON
25	Wheel Speed Sensor Shadow Zone Failure - Front Left	0x50	0x35	0x1D	☺	☺	OFF	ON	ON
26	Wheel Speed Sensor Shadow Zone Failure - Front Right	0x50	0x40	0x1D	☺	☺	OFF	ON	ON
27	Wheel Speed Sensor Shadow Zone Failure - Rear Left	0x50	0x45	0x1D	☺	☺	OFF	ON	ON
28	Wheel Speed Sensor Shadow Zone Failure - Rear Right	0x50	0x50	0x1D	☺	☺	OFF	ON	ON
29	Mismatched Tire	0x52	0x45	0x08	☺	☺	ON	ON	ON
30	Open Solenoid - Iso Front Left	0x50	0x65	0x13	☺	☺	OFF	ON	ON
31	Open Solenoid - Iso Front Right	0x50	0x75	0x13	☺	☺	OFF	ON	ON
32	Open Solenoid - Iso Rear Left	0x50	0x85	0x13	☺	☺	ON	ON	ON
33	Open Solenoid - Iso Rear Right	0x50	0x95	0x13	☺	☺	ON	ON	ON
34	Open Solenoid - Dump Front Left	0x50	0x60	0x13	☺	☺	OFF	ON	ON
35	Open Solenoid - Dump Front Right	0x50	0x70	0x13	☺	☺	OFF	ON	ON
36	Open Solenoid - Dump Rear Left	0x50	0x80	0x13	☺	☺	OFF	ON	ON

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No.	Description	Fault Byte-1	Fault Byte-2	Fault Byte-3	ABS	ESC	EBD Lamp	ABS Lamp	ESC Lamp
37	Open Solenoid - Dump Rear Right	0x50	0x90	0x13	☺	☺	OFF	ON	ON
38	Open Solenoid - TC ISO 1	0x51	0x41	0x13	X	☺	OFF	OFF	ON
39	Open Solenoid - TC ISO 2	0x51	0x51	0x13	X	☺	OFF	OFF	ON
40	Open Solenoid - TC Supply 1	0x51	0x46	0x13	X	☺	OFF	OFF	ON
41	Open Solenoid - TC Supply 2	0x51	0x56	0x13	X	☺	OFF	OFF	ON
42	Shorted Solenoid Driver - Iso Front Left	0x50	0x66	0x12	☺	☺	ON	ON	ON
43	Shorted Solenoid Driver - Iso Front Right	0x50	0x76	0x12	☺	☺	ON	ON	ON
44	Shorted Solenoid Driver - Iso Rear Left	0x50	0x86	0x12	☺	☺	ON	ON	ON
45	Shorted Solenoid Driver - Iso Rear Right	0x50	0x96	0x12	☺	☺	ON	ON	ON
46	Shorted Solenoid Driver - Dump Front Left	0x50	0x61	0x12	☺	☺	ON	ON	ON
47	Shorted Solenoid Driver - Dump Front Right	0x50	0x71	0x12	☺	☺	ON	ON	ON
48	Shorted Solenoid Driver - Dump Rear Left	0x50	0x81	0x12	☺	☺	ON	ON	ON
49	Shorted Solenoid Driver - Dump Rear Right	0x50	0x91	0x12	☺	☺	ON	ON	ON
50	Shorted Solenoid Driver - TC ISO 1	0x51	0x42	0x12	X	☺	ON	ON	ON
51	Shorted Solenoid Driver - TC ISO 2	0x51	0x52	0x12	X	☺	ON	ON	ON
52	Shorted Solenoid Driver - TC Supply 1	0x51	0x47	0x12	X	☺	ON	ON	ON
53	Shorted Solenoid Driver - TC Supply 2	0x51	0x57	0x12	X	☺	ON	ON	ON
54	Leaky driver fault for ISO Front Left solenoid	0x50	0x66	0x12	☺	☺	ON	ON	ON
55	Leaky driver fault for ISO Front Right solenoid	0x50	0x76	0x12	☺	☺	ON	ON	ON
56	Leaky driver fault for ISO Rear Left solenoid	0x50	0x86	0x12	☺	☺	ON	ON	ON
57	Leaky driver fault for ISO Rear Right solenoid	0x50	0x96	0x12	☺	☺	ON	ON	ON
58	Leaky driver fault for Dump Front Left solenoid	0x50	0x61	0x12	☺	☺	ON	ON	ON

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No.	Description	Fault Byte-1	Fault Byte-2	Fault Byte-3	ABS	ESC	EBD Lamp	ABS Lamp	ESC Lamp
59	Leaky driver fault for Dump Front Right solenoid	0x50	0x71	0x12	☺	☺	ON	ON	ON
60	Leaky driver fault for Dump Rear Left solenoid	0x50	0x81	0x12	☺	☺	ON	ON	ON
61	Leaky driver fault for Dump Rear Right solenoid	0x50	0x91	0x12	☺	☺	ON	ON	ON
62	Leaky driver fault for TC ISO1 solenoid	0x51	0x42	0x12	X	☺	ON	ON	ON
63	Leaky driver fault for TC ISO2 solenoid	0x51	0x52	0x12	X	☺	ON	ON	ON
64	Leaky driver fault for TC SUPPLY1 solenoid	0x51	0x47	0x12	X	☺	ON	ON	ON
65	Leaky driver fault for TC SUPPLY2 solenoid	0x51	0x57	0x12	X	☺	ON	ON	ON
66	Leaky Driver Of Unknown Coil	0x50	0x68	0x12	X	☺	ON	ON	ON
67	Shorted Solenoid - Iso Front Left	0x50	0x65	0x12	☺	☺	OFF	ON	ON
68	Shorted Solenoid - Iso Front Right	0x50	0x75	0x12	☺	☺	OFF	ON	ON
69	Shorted Solenoid - Iso Rear Left	0x50	0x85	0x12	☺	☺	ON	ON	ON
70	Shorted Solenoid - Iso Rear Right	0x50	0x95	0x12	☺	☺	ON	ON	ON
71	Shorted Solenoid - Dump Front Left	0x50	0x60	0x12	☺	☺	OFF	ON	ON
72	Shorted Solenoid - Dump Front Right	0x50	0x70	0x12	☺	☺	OFF	ON	ON
73	Shorted Solenoid - Dump Rear Left	0x50	0x80	0x12	☺	☺	OFF	ON	ON
74	Shorted Solenoid - Dump Rear Right	0x50	0x90	0x12	☺	☺	OFF	ON	ON
75	Shorted Solenoid - TC ISO1	0x51	0x41	0x12	X	☺	OFF	OFF	ON
76	Shorted Solenoid - TC ISO2	0x51	0x51	0x12	X	☺	OFF	OFF	ON
77	Shorted Solenoid - TC SUPPLY1	0x51	0x46	0x12	X	☺	OFF	OFF	ON
78	Shorted Solenoid - TC SUPPLY2	0x51	0x56	0x12	X	☺	OFF	OFF	ON
79	Solenoid Overtemperature - Iso Front Left	0x52	0x66	0x4B	☺	☺	OFF	ON	ON
80	Solenoid Overtemperature - Iso Front Right	0x52	0x76	0x4B	☺	☺	OFF	ON	ON

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No.	Description	Fault Byte-1	Fault Byte-2	Fault Byte-3	ABS	ESC	EBD Lamp	ABS Lamp	ESC Lamp
81	Solenoid Overtemperature - Iso Rear Left	0x52	0x86	0x4B	☺	☺	ON	ON	ON
82	Solenoid Overtemperature - Iso Rear Right	0x52	0x96	0x4B	☺	☺	ON	ON	ON
83	Solenoid Overtemperature - Dump Front Left	0x52	0x65	0x4B	☺	☺	OFF	ON	ON
84	Solenoid Overtemperature - Dump Front Right	0x52	0x75	0x4B	☺	☺	OFF	ON	ON
85	Solenoid Overtemperature - Dump Rear Left	0x52	0x85	0x4B	☺	☺	OFF	ON	ON
86	Solenoid Overtemperature - Dump Rear Right	0x52	0x95	0x4B	☺	☺	OFF	ON	ON
87	Solenoid Overtemperature - TC ISO1	0x52	0x41	0x4B	X	☺	OFF	OFF	ON
88	Solenoid Overtemperature - TC ISO2	0x52	0x51	0x4B	X	☺	OFF	OFF	ON
89	Solenoid Overtemperature - TC Supply1	0x52	0x42	0x4B	X	☺	OFF	OFF	ON
90	Solenoid Overtemperature - TC Supply2	0x52	0x52	0x4B	X	☺	OFF	OFF	ON
91	Loss Of Solenoid Ground In The Wiring Harness	0x51	0x09	0x00	☺	☺	OFF	ON	ON
92	ABS Open Pump Motor	0x51	0x10	0x13	☺	☺	OFF	ON	ON
93	ABS Pump Motor Supply Open	0x51	0x11	0x00	☺	☺	OFF	ON	ON
94	ABS Pump Motor Driver Open	0x51	0x11	0x13	☺	☺	ON	ON	ON
95	ABS Locked Pump Motor	0x51	0x10	0x71	☺	☺	OFF	ON	ON
96	LBIST Fault R4	0x53	0x50	0x00	X	☺	ON	ON	ON
97	LBIST Fault M3	0x53	0x50	0x00	☺	☺	ON	ON	ON
98	MISR Fault R4	0x53	0x50	0x00	X	☺	ON	ON	ON
99	MISR Fault M3	0x53	0x50	0x00	X	☺	ON	ON	ON
100	Dual Clock Compare	0x53	0x50	0x00	X	☺	ON	ON	ON
101	IMM Fault R4	0x53	0x50	0x00	X	☺	ON	ON	ON
102	IMM Fault M3	0x53	0x50	0x00	X	☺	ON	ON	ON

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No.	Description	Fault Byte-1	Fault Byte-2	Fault Byte-3	ABS	ESC	EBD Lamp	ABS Lamp	ESC Lamp
103	Lockup M3	0x53	0x50	0x00	X	☺	ON	ON	ON
104	Imprecise Abort M3	0x53	0x50	0x00	☺	☺	ON	ON	ON
105	ROM Check Failure_R4	0x53	0x50	0x00	X	☺	ON	ON	ON
106	ROM Check Failure_M3	0x53	0x50	0x00	☺	☺	ON	ON	ON
107	CRC ROM Check Failure_R4	0x53	0x50	0x00	X	☺	ON	ON	ON
108	CRC ROM Check Failure_M3	0x53	0x50	0x00	☺	☺	ON	ON	ON
109	dynamic RAM Check Failure R4	0x53	0x50	0x00	☺	☺	ON	ON	ON
110	RAM Check Failure_R4	0x53	0x50	0x00	X	☺	ON	ON	ON
111	RAM Check Failure_M3	0x53	0x50	0x00	☺	☺	ON	ON	ON
112	RAM ECC Check Failure_M3	0x53	0x50	0x00	☺	☺	ON	ON	ON
113	DMA Failure	0x53	0x50	0x00	X	☺	ON	ON	ON
114	RAM corrupts associated with the state variables M3	0x53	0x50	0x00	☺	☺	ON	ON	ON
115	RAM corrupts associated with the state variables R4	0x53	0x50	0x00	X	☺	ON	ON	ON
116	RAM Check Failure HET	0x53	0x50	0x00	☺	☺	ON	ON	ON
117	RAM Check Failure NHET	0x53	0x50	0x00	X	☺	ON	ON	ON
118	RAM Check Failure NHET PARITY	0x53	0x50	0x00	X	☺	ON	ON	ON
119	HET parity error	0x53	0x50	0x00	☺	X	ON	ON	ON
120	Cpu Mode Stack Fault R4	0x53	0x50	0x00	X	☺	ON	ON	ON
121	Cpu Mode Stack Fault M3	0x53	0x50	0x00	☺	☺	ON	ON	ON
122	Task Stack Fault R4	0x53	0x50	0x00	X	☺	ON	ON	ON
123	Task Stack Fault M3	0x53	0x50	0x00	☺	☺	ON	ON	ON
124	Task Overrun Failure R4	0x53	0x80	0x00	X	☺	ON	ON	ON

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No.	Description	Fault Byte-1	Fault Byte-2	Fault Byte-3	ABS	ESC	EBD Lamp	ABS Lamp	ESC Lamp
125	Task Overrun Failure M3	0x53	0x80	0x00	☺	☺	ON	ON	ON
126	External Watchdog Failure	0x53	0x50	0x47	☺	☺	ON	ON	ON
127	External Secondary Shutoff Path Failure (WDEN pin)	0x53	0x50	0x47	X	☺	ON	ON	ON
128	Watchdog Dynamic Test Failure	0x53	0x50	0x47	☺	☺	ON	ON	ON
129	Digital Watchdog Failure	0x53	0x50	0x47	☺	X	ON	ON	ON
130	AD Peripheral Timeout Failure	0x53	0x50	0x00	☺	☺	ON	ON	ON
131	Lockup of ADC Event Group	0x53	0x50	0x00	X	☺	ON	ON	ON
132	MIRAM AD Refresh Fault	0x53	0x50	0x00	☺	☺	ON	ON	ON
133	MIRAM DIAG Register Fault	0x53	0x50	0x00	X	☺	ON	ON	ON
134	Miram VBAT overcurrent	0x53	0x50	0x00	☺	X	ON	ON	ON
135	Miram Boost Over Voltage	0x53	0x50	0x00	☺	X	ON	ON	ON
136	Miram Boost Under Voltage	0x53	0x50	0x00	☺	X	ON	ON	ON
137	Miram Select Failure	0x53	0x50	0x00	☺	X	ON	ON	ON
138	Miram Pre-Register Mode Select Failure	0x53	0x50	0x00	☺	X	ON	ON	ON
139	Miram U5 failure	0x53	0x50	0x00	☺	X	ON	ON	ON
140	RTOS Failure R4	0x53	0x73	0x00	X	☺	ON	ON	ON
141	RTOS Failure M3	0x53	0x73	0x00	☺	☺	ON	ON	ON
142	Configuration Failure M3 FM	0x53	0x50	0x00	X	☺	ON	ON	ON
143	Unimplemented Interrupt R4	0x53	0x50	0x00	X	☺	ON	ON	ON
144	Unimplemented Interrupt M3	0x53	0x50	0x00	☺	☺	ON	ON	ON
145	Unexpected Exception R4	0x53	0x81	0x00	X	☺	ON	ON	ON
146	Unexpected Exception M3	0x53	0x81	0x00	☺	☺	ON	ON	ON

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No.	Description	Fault Byte-1	Fault Byte-2	Fault Byte-3	ABS	ESC	EBD Lamp	ABS Lamp	ESC Lamp
147	Solenoid Timeout Failure	0x53	0x50	0x00	☺	☺	ON	ON	ON
148	Nhet Interrupt Test Failure	0x53	0x70	0x04	X	☺	ON	ON	ON
149	Het Interrupt Test Failure	0x53	0x70	0x04	☺	☺	ON	ON	ON
150	NHet Periodic Interrupt Failure	0x53	0x70	0x04	X	☺	ON	ON	ON
151	Het Periodic Interrupt Failure	0x53	0x70	0x04	☺	☺	ON	ON	ON
152	NHet Watchdog Timeout	0x53	0x72	0x04	X	☺	ON	ON	ON
153	Het Watchdog Timeout	0x53	0x72	0x04	☺	☺	ON	ON	ON
154	NHet Program Error	0x53	0x71	0x04	X	☺	ON	ON	ON
155	Het Program Error	0x53	0x71	0x04	☺	☺	ON	ON	ON
156	NHet Program Overflow	0x53	0x73	0x04	X	☺	ON	ON	ON
157	Het Program Overflow	0x53	0x73	0x04	☺	☺	ON	ON	ON
158	NHet Register failure	0x53	0x50	0x00	X	☺	ON	ON	ON
159	Het Register failure	0x53	0x50	0x00	☺	☺	ON	ON	ON
160	Mismatched velocity calculation between both core	0x53	0x50	0x00	☺	☺	ON	ON	ON
161	FSMC communication failure M3	0x53	0x50	0x00	X	☺	ON	ON	ON
162	FSMC communication failure R4	0x53	0x50	0x00	X	☺	ON	ON	ON
163	FM communication failure M3	0x53	0x50	0x00	X	☺	ON	ON	ON
164	FM communication failure R4	0x53	0x50	0x00	X	☺	ON	ON	ON
165	Ignition Voltage Low	0x53	0x00	0x16	☺	☺	OFF	ON	ON
166	Ignition Voltage High	0x53	0x00	0x17	☺	☺	OFF	ON	ON
167	Ignition Voltage Excessively Low	0x53	0x00	0x16	☺	☺	OFF	ON	ON

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No.	Description	Fault Byte-1	Fault Byte-2	Fault Byte-3	ABS	ESC	EBD Lamp	ABS Lamp	ESC Lamp
168	Ignition Voltage Excessively High	0x53	0x00	0x17	☺	☺	OFF	ON	ON
169	Ignition Voltage Erratic Low	0x53	0x00	0x2F	☺	☺	OFF	ON	ON
170	Ignition Voltage Erratic High	0x53	0x00	0x2F	☺	☺	OFF	ON	ON
171	Supply voltage erratic Pressure Sensor	0x51	0x37	0x00	X	☺	OFF	OFF	ON
172	Internal 5 V Sensor Supply failure	0x53	0x50	0x1C	☺	☺	ON	ON	ON
173	µC ADC refernces fault	0x53	0x50	0x1C	X	☺	ON	ON	ON
174	SPI Failure	0x53	0x75	0x00	☺	☺	ON	ON	ON
175	NVRAM multiple write failures	0x53	0x50	0x00	☺	☺	ON	ON	ON
176	NVRAM Chip not operational	0x53	0x50	0x00	☺	☺	ON	ON	ON
177	NVRAM Reset Ongoing	0x53	0x50	0x00	☺	X	ON	ON	ON
178	Open power switch for slip control	0x51	0x21	0x13	☺	☺	ON	ON	ON
179	Shorted power switch for slip control	0x51	0x21	0x12	☺	☺	OFF	ON	ON
180	Vehicle deceleration stronger than allowed deceleration	0x51	0x18	0x17	X	☺	OFF	OFF	OFF
181	Vehicle deceleration lower than allowed deceleration.	0x51	0x18	0x17	X	☺	OFF	OFF	OFF
182	Vehicle Speed Output Overcurrent Failure - channel	0x51	0x19	0x17	X	☺	OFF	OFF	OFF
183	Steering Angle Sensor Fault in last ignition cycle	0x51	0x12	0x00	X	☺	OFF	OFF	ON
184	Steering wheel angle Battery change	0x52	0x12	0x02	X	☺	OFF	OFF	ON
185	Steering wheel angle offset error	0x53	0x12	0x08	X	☺	OFF	OFF	ON
186	Steering wheel angle raw offset error	0x53	0x12	0x08	X	☺	OFF	OFF	ON
187	Steering wheel angle range error	0x51	0x12	0x00	X	☺	OFF	OFF	ON
188	Steering wheel not alive error	0x51	0x12	0x00	X	☺	OFF	OFF	ON
189	Steering wheel angle sensor missing calibration	0x50	0x12	0x08	X	☺	OFF	OFF	ON

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No.	Description	Fault Byte-1	Fault Byte-2	Fault Byte-3	ABS	ESC	EBD Lamp	ABS Lamp	ESC Lamp
190	Steering wheel angle step error	0x53	0x12	0x08	X	☺	OFF	OFF	ON
191	YAW Sensor Shorted High Failure	0x52	0x15	0x12	X	☺	OFF	OFF	ON
192	YAW Sensor Open Or Short To Ground Failure	0x52	0x15	0x14	X	☺	OFF	OFF	ON
193	YAW Sensor Erratic Failure	0x52	0x15	0x2F	X	☺	OFF	OFF	ON
194	Yaw Sensor Diag state	0x52	0x15	0x16	X	☺	OFF	OFF	ON
195	YAW Sensor Not Alive Error	0x52	0x15	0x16	X	☺	OFF	OFF	ON
196	Yaw Sensor Ground loss	0x52	0x15	0x16	X	☺	OFF	OFF	ON
197	Angular Rate Sensor Configuration Failure	0x52	0x15	0x16	X	☺	OFF	OFF	ON
198	Angular Rate Sensor Built-in Test Failure	0x52	0x15	0x16	X	☺	OFF	OFF	ON
199	YAW Sensor Raw Offest Failure	0x52	0x25	0x08	X	☺	OFF	OFF	ON
200	YAW Sensor Offest Failure	0x52	0x25	0x08	X	☺	OFF	OFF	ON
201	YAW Sensor Gain Error	0x53	0x15	0x08	X	☺	OFF	OFF	ON
202	YAW Sensor Missing Calibration	0x53	0x63	0x54	X	☺	OFF	OFF	ON
203	Lateral Sensor Out Of Range	0x52	0x15	0x08	X	☺	OFF	OFF	ON
204	Lateral Acceleration Sensor Offest Failure	0x52	0x15	0x08	X	☺	OFF	OFF	ON
205	Lateral Acceleration Sensor Raw Offest Failure	0x52	0x15	0x08	X	☺	OFF	OFF	ON
206	Lateral Acceleration Sensor Gain Error	0x53	0x25	0x08	X	☺	OFF	OFF	ON
207	Lateral sensor not alive error	0x52	0x15	0x08	X	☺	OFF	OFF	ON
208	Lateral sensor missing calibration	0x53	0x61	0x54	X	☺	OFF	OFF	ON
209	Acceleration Sensor over Temp Failure	0x51	0x15	0x4B	X	☺	OFF	OFF	ON
210	Accel Sensor Configuration Failure	0x51	0x15	0x16	X	☺	OFF	OFF	ON
211	Acceleration Sensor Power On Reset	0x51	0x15	0x16	X	☺	OFF	OFF	ON

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No.	Description	Fault Byte-1	Fault Byte-2	Fault Byte-3	ABS	ESC	EBD Lamp	ABS Lamp	ESC Lamp
212	Acceleration Sensor Diag status Test Failure	0x51	0x15	0x16	X	☺	OFF	OFF	ON
213	Acceleration Sensor SPI not operational	0x51	0x15	0x16	X	☺	OFF	OFF	ON
214	MC Pressure Sensor Ohmic Open or Shorted High	0x51	0x35	0x12	X	☺	OFF	OFF	ON
215	MC Pressure Sensor Shorted Low	0x51	0x34	0x14	X	☺	OFF	OFF	ON
216	MC Pressure Sensor Correlation Failure	0x51	0x30	0x00	X	☺	OFF	OFF	ON
217	MC Pressure Sensor Erratic	0x51	0x26	0x2F	X	☺	OFF	OFF	ON
218	Pressure Sensor Offset Error	0x51	0x33	0x00	X	☺	OFF	OFF	ON
219	Pressure Sensor Raw Offset Error	0x51	0x33	0x00	X	☺	OFF	OFF	ON
220	Pressure Sensor Not Alive Error	0x51	0x29	0x00	X	☺	OFF	OFF	ON
221	Inertial Sensor Initialization Fault	0x51	0x15	0x29	X	☺	OFF	OFF	ON
222	Acceleration Sensor is satureated	0x51	0x15	0x29	X	☺	OFF	OFF	ON
223	Vacuum Chamber Shorted to Ground	0x51	0x23	0x16	X	☺	OFF	OFF	OFF
224	Vacuum Chamber Open or Shorted to Supply	0x51	0x23	0x17	X	☺	OFF	OFF	OFF
225	Vacuum Chamber Signal Erratic	0x51	0x23	0x2F	X	☺	OFF	OFF	OFF
226	Vacuum Chamber Sensor Step Failure	0x51	0x23	0x08	X	☺	OFF	OFF	OFF
227	Vacuum Chamber not alive error	0x51	0x23	0x00	X	☺	OFF	OFF	OFF
228	Vacuum Sensor Vacuum Chamber Correlation Failure	0x51	0x23	0x00	X	☺	OFF	OFF	OFF
229	Vacuum Sensor Vaccum Chamber RAW Correlation Failure	0x51	0x23	0x00	X	☺	OFF	OFF	OFF
230	Vacuum Sensor Supply Volt Failure	0x51	0x23	0x02	X	☺	OFF	OFF	OFF
231	Defective Vacuum Hose	0x53	0x23	0x08	X	☺	OFF	OFF	OFF
232	Reverse Switch Failure	0x51	0x34	0x08	X	☺	OFF	OFF	OFF
233	Brake Switch Always Active	0x51	0x61	0x00	☺	☺	OFF	OFF	OFF

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No.	Description	Fault Byte-1	Fault Byte-2	Fault Byte-3	ABS	ESC	EBD Lamp	ABS Lamp	ESC Lamp
234	Brake Switch Never Active	0x51	0x61	0x00	☺	☺	OFF	OFF	OFF
235	Driver Mode Switch Stuck ON	0x51	0x62	0x00	X	☺	OFF	OFF	OFF
236	Driver Mode Switch Invalid	0x51	0x62	0x00	X	☺	OFF	OFF	OFF
237	System Configuration (Calibration) Error	0x52	0x13	0x54	X	☺	OFF	OFF	ON
238	System Configuration (Calibration) not programmed	0x52	0x12	0x55	X	☺	OFF	OFF	ON
239	CAN Bus 0 Off	0x52	0x01	0x88	☺	☺	OFF	OFF	ON
240	CAN Bus 0 Mute	0x52	0x01	0x08	X	☺	OFF	OFF	ON
241	CAN Bus 0 Transmission Acknowledge	0x52	0x01	0x08	☺	X	OFF	OFF	ON
242	CAN Bus 0 Over Run	0x52	0x01	0x88	☺	☺	OFF	OFF	ON
243	CAN Bus 0 Hardware INIT	0x53	0x50	0x00	☺	☺	ON	ON	ON
244	CAN Bus 0 RAM Check Fault	0x53	0x50	0x00	☺	X	ON	ON	ON
245	Missing Engine Controller Message EMS_205	0xE1	0x00	0x87	X	☺	OFF	OFF	ON
246	Missing Engine Controller Message EMS_255	0xE1	0x00	0x87	☺	☺	OFF	OFF	ON
247	Missing Engine Controller Message EMS_275	0xE1	0x00	0x87	X	☺	OFF	OFF	ON
248	Missing Engine Controller Message EMS_265	0xE1	0x00	0x87	☺	☺	OFF	OFF	ON
249	Missing Engine Controller Message EMS_355	0xE1	0x00	0x87	☺	☺	OFF	OFF	ON
250	Missing Gearbox Message TCU_238	0xE1	0x01	0x87	X	☺	OFF	OFF	OFF
251	Missing Gearbox Message TCU_268	0xE1	0x01	0x87	X	☺	OFF	OFF	OFF
252	Missing Steering Message SAS_180	0xE1	0x12	0x04	X	☺	OFF	OFF	ON
253	Missing SRS Controller Message SRS_050	0xE1	0x03	0x87	X	☺	OFF	OFF	OFF
254	Missing IP Controller Message IP_270	0xE1	0x04	0x87	X	☺	OFF	OFF	OFF
255	Missing EPB Controller Message EPB_256	0xE1	0x14	0x87	X	☺	OFF	OFF	OFF

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No.	Description	Fault Byte-1	Fault Byte-2	Fault Byte-3	ABS	ESC	EBD Lamp	ABS Lamp	ESC Lamp
256	Missing EPB Controller Message EPB_246	0xE1	0x14	0x87	X	☺	OFF	OFF	OFF
257	Engine Drive Torque Invalid	0x51	0x01	0x81	X	☺	OFF	OFF	ON
258	Engine Frictional Torque Invalid	0x51	0x01	0x81	X	☺	OFF	OFF	ON
259	Engine Real Torque Invalid	0x51	0x01	0x81	X	☺	OFF	OFF	ON
260	Engine Speed Invalid	0x51	0x01	0x81	X	☺	OFF	OFF	ON
261	Engine Brake Pedal Invalid (ESC) & Brake Switch Failure Hardwired (ABS)	0x51	0x01	0x81	☺	☺	OFF	OFF	ON
262	Engine Accel Pedal Invalid	0x51	0x01	0x81	☺	☺	OFF	OFF	ON
263	Engine Cool Temperature Invalid	0x51	0x01	0x81	X	☺	OFF	OFF	ON
264	Engine Total Odometer Invalid	0x51	0x01	0x81	X	☺	OFF	OFF	ON
265	EMS 265 Message counter invalid fault	0x51	0x01	0x81	X	☺	OFF	OFF	ON
266	EMS Manifold Pressure Invalid	0x51	0x01	0x81	X	☺	OFF	OFF	ON
267	EMS Altitude Factor Invalid	0x51	0x01	0x81	X	☺	OFF	OFF	ON
268	EMS clutch pedal invalid fault.	0x51	0x01	0x81	X	☺	OFF	OFF	ON
269	Engine Maximum Torque Invalid	0x51	0x01	0x81	X	☺	OFF	OFF	ON
270	Engine Minimum Torque Invalid	0x51	0x01	0x81	X	☺	OFF	OFF	ON
271	Engine Torque Constant Invalid	0x51	0x01	0x81	X	☺	OFF	OFF	ON
272	Gear Position Shift Progress Invalid	0x51	0x02	0x81	X	☺	OFF	OFF	ON
273	Actual Gear Invalid	0x51	0x02	0x81	X	☺	OFF	OFF	ON
274	Target Gear Invalid	0x51	0x02	0x81	X	☺	OFF	OFF	ON
275	Steering Angle Invalid	0x51	0x12	0x00	X	☺	OFF	OFF	ON
276	Steering Angle Speed Invalid	0x51	0x12	0x00	X	☺	OFF	OFF	ON
277	Steering Angle Sensor Batttery Warning Invalid	0x51	0x12	0x00	X	☺	OFF	OFF	ON

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No.	Description	Fault Byte-1	Fault Byte-2	Fault Byte-3	ABS	ESC	EBD Lamp	ABS Lamp	ESC Lamp
278	Steering Angle Sensor Invalid	0x51	0x12	0x00	X	☺	OFF	OFF	ON
279	Steering Angle Sensor Message Counter Invalid	0x51	0x12	0x00	X	☺	OFF	OFF	ON
280	Steering Angle Sensor checksum Invalid	0x51	0x12	0x00	X	☺	OFF	OFF	ON
281	Driver Door Status Invalid	0x51	0x22	0x81	X	☺	OFF	OFF	OFF
282	Controlled System Shutdown Failure	0x53	0x78	0x04	☺	☺	OFF	OFF	ON
283	EBD ONLY MODE (For developing ECU)	0x5E	0x06	0x00	☺	☺	OFF	ON	ON
284	Longitudinal Sensor Out Of Range	0x52	0x16	0x08	X	☺	OFF	OFF	OFF
285	Longitudinal Acceleration Sensor Offest Failure	0x52	0x16	0x08	X	☺	OFF	OFF	OFF
286	Longitudinal Acceleration Sensor Raw Offest Failure	0x52	0x16	0x08	X	☺	OFF	OFF	OFF
287	Longitudinal Acceleration Sensor Gain Error	0x53	0x16	0x08	X	☺	OFF	OFF	OFF
288	Long sensor output not moving	0x52	0x16	0x08	X	☺	OFF	OFF	OFF
289	Long Sensor Missing Calibration	0x53	0x64	0x54	X	☺	OFF	OFF	OFF
290	**EPB Failure:**								
291	EPB 256 Checksum Fault	0x51	0x14	0x00	X	☺	OFF	OFF	OFF
292	EPB 246 Checksum Fault	0x51	0x14	0x00	X	☺	OFF	OFF	OFF
293	EPB 246 Message counter invalid fault	0x51	0x14	0x81	X	☺	OFF	OFF	OFF
294	EPB 256 message counter invalid	0x51	0x14	0x81	X	☺	OFF	OFF	OFF
295	EPB Deceleration request invalid	0x51	0x14	0x81	X	☺	OFF	OFF	OFF
296	EPB switch position invalid	0x51	0x14	0x81	X	☺	OFF	OFF	OFF
297	EPB Clamp force invalid fault.	0x51	0x14	0x81	X	☺	OFF	OFF	OFF
298	EPB Failed	0x51	0x14	0x81	X	☺	OFF	OFF	OFF
299	Long Acceleration from EPB is invalid	0x51	0x14	0x81	X	☺	OFF	OFF	OFF

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No.	Description	Fault Byte-1	Fault Byte-2	Fault Byte-3	ABS	ESC	EBD Lamp	ABS Lamp	ESC Lamp
300	BHM-APC learned offset value is excessive out of range	0x51	0x17	0x00	X	☺	OFF	OFF	OFF
301	BHM-APC has detected a defect pump, TC-Iso or TC-Supply	0x51	0x17	0x00	X	☺	OFF	OFF	OFF
302	Exc. No. of failed learning attempts by press. apply timeout	0x51	0x17	0x00	X	☺	OFF	OFF	OFF
303	Exc. No. of failed learning attempts by pressure spikes	0x51	0x17	0x00	X	☺	OFF	OFF	OFF
304	Exc. No. of failed learning attempts by too less rpm of pump	0x51	0x17	0x00	X	☺	OFF	OFF	OFF
305	Exc. No. of failed learning attempts by input signal quality	0x51	0x17	0x00	X	☺	OFF	OFF	OFF
306	Auto Hold Switch Stuck ON	0x51	0x63	0x00	X	☺	OFF	OFF	OFF
307	Auto Hold Switch Invalid	0x51	0x63	0x00	X	☺	OFF	OFF	OFF
308	SSM_NO_FEEDBACK_PB_APPLIED	0x51	0x64	0x00	X	☺	OFF	OFF	OFF

Note:

☺: This DTC is implemented in ABS or ESC.

X: This DTC is not implemented in ABS or ESC.

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5.2. Supported Snapshot Data

5.2.1. Mandatory Snapshot data

The following identifiers are to be supported by all ECUs.

Item	DataIdentifier	Number of Bytes	Resolution
Supply Voltage	0112	1	0.1
Odometer Reading	E101	3	1km
Ignition Status	D001	1	

Table 5-1: Mandatory Snapshot Data

5.2.2. Supported Additional Snapshot Identifiers

Item	DataIdentifier	Number of Bytes	Resolution
Vehicle speed	010F	1	1 bit = 2 km/hr
System Status	D002	1	
Fault Id	D003	2	

Table 5-2: Additional Snapshot Data

5.2.3. Extended Snapshot Information

Item	DataIdentifier	Number of Bytes	Resolution
Ageing Counter	D004	1	

Table 5-3: Extended Snapshot Data

NOTE:

The Snapshot Data for a DTC gets cleared if we clear the diagnostics using SID$14 or after 40 ignition cycles.

5.2.4. Description of snapshot data

Item	Bit	Bit Description	0	1
Ignition Status	7	Reserved	NA	NA
	6	Global Engine Running Status	Off	On

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	5	Global KL.15 (Ignition Switch)	Off	On
	4	Reserved	NA	NA
	3	Reserved	NA	NA
	2	System Engine Running Status	Off	On
	1	System KL.15 (Ignition Switch) HW I/O port status	Off	On
	0	Reserved	NA	NA

Table 5-4: Ignition status

Item	Bit	Bit Description	0	1
System Status	7	Reserved Default 0	0	
	6	Reserved Default 0	0	
	5	ESC enable status	Disabled	Enabled
	4	ESC Mode	Passive	Active
	3	TC Brake Intervention	Passive	Active
	2	TC Engine Intervention	Passive	Active
	1	ABS Mode	Passive	Active
	0	EBD Mode	Passive	Active

Table 5-5: System status

Item	description
Ageing counter	If the fault is not observed in next 40 ignition cycles it will be removed from the history (Aging counter gets incremented by 1 for every ignition cycle. Maximum value is 40)

Table 5-6: Ageing counter

5.3. DTC format

According to ISO 15031-6 and ISO 14229-1 DTC format, the DTC defined in this specification

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consists of 4 bytes, see table 1:

Table 1 DTC information format

DTC information format			
DTCHighByte(Hex)	DTCMiddleByte(Hex)	DTCLowByte(Hex)	StatusOfDTC

In which DTCHighByte，DTCMiddleByte，DTCLowByte indicate the diagnostic fault codes of the service; StatusOfDTC indicates the status of the fault code.

DTCHighByte and DTCMiddleByte indicate the internal fault codes, the corresponding 5 parts standard fault codes are defined in Table 2:

Table 2 Internal fault codes and the corresponding 5 parts standard fault codes (1)

Internal Fault Codes	**DTCHighByte(Hex)**								**DTCMiddleByte(Hex)**							
	Bit 15	Bit 14	Bit 13	Bit 12	bit 11	Bit 10	Bit 9	Bit 8	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	bit 0
5 parts standard fault codes	Part 1		Part 2		Part 3				Part 4 and Part 5							

In 5 parts standard fault codes, the Part 1 is the character and the following 4 parts are the number, for example P0120。

The first character indicates what system the fault codes belong to: “P” Powertrain system, “B” Body, “C” Chassis，“U” Network，see Table 3:

Table 3 First part of the 5 parts standard fault codes

Internal fault codes Bit15 Bit14	**Part 1 of the standard fault codes**	**System**
00	P	Powertrain
01	C	Chassis
10	B	Body
11	U	Network

The second part (Number 0、1、2 or 3) indicates the fault type: “0” means the general fault codes defined by SAE; “1” means the extended fault codes defined by the vehicle manufacturer; “2” or “3” is reserved, see Table 4:

Table 4 The second part of the 5 parts fault codes

Internal fault codes Bit13 Bit12	**Part 2 of the standard codes**	**Fault type**
00	0	Fault codes defined by ISO/SAE
01	1	Fault codes defined by the vehicle manufacturer
10	2	Fault codes defined by ISO/SAE

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11	3	Reserved by ISO/SAE

The third part means what subsystem the fault codes belong to.

The last two parts indicate the detailed fault object and fault type.

DTCLowByte describes the fault type and subtype, which accords with ISO 15031-6. For the DTC which doesn't need this byte information, it will be padded with 0x00.

StatusOfDTC indicates of the DTC status. See Annex C for the details.

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6. Annex

6.1. Annex A: Data definition for diagnostic service negative response codes (NRC_)

Table A Negative response codes (NRC_) data definition

Data	Description	Abbreviation
0~0x0F	reserved	—
0x10	generalReject If the negative response codes don't belong to the other items in this table, the server will reject providing the service to the client.	GR
0x11	serviceNotSupported The server doesn't support the diagnostic request service from the client.	SNS
0x12	subfuntionNotSupported The server doesn't support the request sub-function from the client.	SFNS
0x13	incorrectMessageLengthOrInvalidFormat The server finds the length (or the format) of the request data from the client doesn't accord to this specification.	IMLOIF
0x14~0x20	reserved	—
0x21	busyRepeatRequest The server is busy and can not handle the request from the client. This negative response code indicates the ending of the diagnostic service.	BRR
0x22	conditionsNotCorrect The conditions for the server to excute the diagnostic service can not be available.	CNC
0x23	reserved	—
0x24	requestSequenceError The server finds the diagnostic service request (or execution) order is not correct.	RSE
0x25~0x30	reserved	—
0x31	requestOutOfRange The server has not the data requested by the client. This negative response code is fit for the servers which support the data writing and reading or the data adjusting.	ROOR
0x32	reserved	—

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Data	Description	Abbreviation
0x33	securityAccessDenied The server prevents the client from doing the diagnostic requests which are inhibited. The reasons are: ● The test conditions of the server can not be met. ● The security status of the server is locked.	SAD
0x34	reserved	—
0x35	invalidKey The server finds the Key returned from the client is not correct.	IK
0x36	exceedNumberOfAttempts The server thinks the security access attempts number of the client exceeds the specified value.	ENOA
0x37	requiredTimeDelayNotExpired The server refuses the security access request from the client because the allowed timing for the server to receive the request is not coming yet.	RTDNE
0x38~0x4F	Reserved for ISO15764	—
0x50~0x6F	Reserved	—
0x70	uploadDownloadNotAccepted The server refuses the Upload/Download operation for the memory of the server from the client due to some failure.	UDNA
0x71	transferDataSuspended The server has stopped the data transferring due to some failure.	TDS
0x72	generalProgrammingFailure During the period of erasing or writing NVRAM, the server stops the diagnostic services due to finding the failure.	GPF
0x73	wrongBlockSequenceCounter The server finds the Block Sequence Counter of the Transmitting Data request data (SID = 0x36) from the client is not correct.	WBSC
0x74~0x77	Reserved	—
0x78	requestCorrectlyReceived-ResponsePending The server has received the request from the client correctly, but handling the request is still onging. The transmitting timing for this negative response code should accord with	RCRRP

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Data	Description	Abbreviation
	$P2_{CAN_Server}$ request. The server should repeat sending this negative response code until handling the request is done.	
0x79~0x7D	Reserved	—
0x7E	subFunctionNotSupportedInActiveSession In current diagnostic mode, the server doesn't support the subfunction of the request service from the client.	SFNSIAS
0x7F	serviceNotSupportedInActiveSession In current diagnostic mode, the server doesn't support the service request from the client.	SNSIAS
0x80	Reserved	—
0x81	rpmTooHigh The server finds the engine can not meet the maximal engine speed condition requirement.	RPMRH
0x82	rpmTooLow The server finds the engine can not meet the minimum engine speed condition requirement.	RPMTL
0x83	engineIsRunning The server can not run the testing because the engine is running. This negative response code is different from NRC = 0x81.	EIR
0x84	engineIsNotRunning The server can not run the testing during the execution period because the engine running is stopped. This negative response code is different from NRC = 0x82.	EINR
0x85	engineRunTimeTooLow The server thinks the engine running time requirement can not be met.	ERTTL
0x86	temperatureTooHigh The server thinks the engine temperature is too high.	TEMPTH
0x87	temperatureTooLow The server thinks the engine temperature is too low.	TEMPTL
0x88	vehicleSpeedTooHigh The server thinks the vehicle speed is too high.	VSTH
0x89	vehicleSpeedTooLow The server thinks the vehicle speed is too low.	VSTL
0x8A	throttle/PedalTooHigh	TPTH

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Data	Description	Abbreviation
	The server thinks the engine throttle position or gas pedal position is too high.	
0x8B	throttle/PedalTooLow The server thinks the engine throttle position or gas pedal position is too low.	TPTL
0x8C	transmissionRangeNotInNeutral The server thinks the transmission can not meet the requirement for the Neutral.	TRNIN
0x8D	transmissionRangeNotInGear The server thinks the transmission can not meet the requirement in Gear range.	TRNIG
0x8E	Reserved	—
0x8F	brakeSwitch(es)NotClosed The server thinks the brake pedal should be pressed for some testing for the sake of the safety.	BSNC
0x90	shifterLeverNotInPark The server thinks the shifter lever should be always pulled up for the sake of the safety.	SLNIP
0x91	torqueConverterClutchLocked The server thinks the torque converter clutch doesn't meet the requirement (Locked).	TCCL
0x92	voltageTooHigh The server thinks the battery voltage is too high.	VTH
0x93	voltageTooLow The server thinks the battery voltage is too low.	VTL
0x94~0xFF	Reserved.	—

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6.2. Annex B: DTC group and DTC Data range definition

As per CHANA DTC request, ABS/ESP DTCs range is: C1000 – C13FF;
ABS/ESP Network DTCs range: U2100 – U217F

6.3. Annex C: DTCStatusMask and statusOfDTC bits definition

As discussed with CHANA, TRW ABS/ESP should support the following DTC state based on TRW internal fault detection mechanism:

Bit 1: Test failed this monitoring cycle
Bit 2: Pending DTC
Bit 3: Confirmed DTC
Bit 6: Test not completed this monitoring cycle
Bit 7: Warning indicator requested.

6.4. Annex D: InputOutputControlParameter definition

Table D InputOutputControlParameter definition

Hex	Description	Cvt	Abbreviaton
00	**returnControlToECU** This value indicates to the server that the client side will not control the input signal any more; The byte number of the controlState in request data: 0; The byte number of the controlState in the positive response: Dependant on dataIdentifier.	U	RCTECU
01	**resetToDefault** The server is recovered to the default state of all the signals and parameters; The byte number of the controlState in request data: 0; The byte number of the controlState in positive response: Dependant on dataIdentifier.	U	RTD
02	**freezeCurrentState** Request to freeze the current state of the input signals, internal parameters or output singals of the server (by inputOutputIdentifier); The byte number of the controlState in the request data: 0; The byte number of the controlState in the positive	U	FCS

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Hex	Description	Cvt	Abbreviaton
	response: Dependant on dataIdentifier.		
03	**shortTermAdjustment** Request to modify the values of the input signals, internal parameters or output singals of the server into the values defined by the parameters of the controlOption (by inputOutputLocalIdentifier); The byte number of the controlState in the request data: Dependant on dataIdentifier; The byte number of the controlState in the positive response: Dependant on dataIdentifier.	U	STA
04-FF	**Reserved**	M	Reserved

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6.5. Annex E: Non-FBL node diagnostic requirement

For non-FBL node, the required services are used for supporting the other FBL node of the CAN network to do the programming download.

The following diagnostic services are required for non-FBL node to support the other FBL nodes to do the programming download:

SID	Diagnostic Service Name	Diagnostic Mode			Supported Addressing		Supported sub-funcations	
		Default	Non-default					
			Programming	Extended	Functional	Physical		
0x10	Diagnostic SessionControl	√	-	√	√	√	01 DefaultSession	√
							02 ProgrammSession	-
							03 ExtendedSession	√
0x28	CommunicationControl	-	-	√	√	√	00 EnableRxAndTx	√
							03 DisableRxAndTx	√
0x85	ControlDTCSetting	-	-	√	√	√	01 ON	√
							02 OFF	√
0x3E	TesterPresent	√	-	√	√	√	00 ZeroSubFunction	√
Note: “√” means Supported, “-” means Not Supported.								

---------- **End of Document** ----------

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